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From perceived attractiveness to visitation: Multi-scale measurement and influencing factors across 143 Chinese historic cities

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ABSTRACT

Effective tourism management requires understanding how attraction attributes and supporting factors shape tourists' perceived attractiveness, and how these in turn translate into visitation. We develop a tourist-centered framework using 610,136 reviews across 2128 heritage attractions in 143 cities, measuring multi-scale perceived attractiveness with a large language model pipeline and visitation with review volume, then identifying the factors that influence both metrics using XGBoost and SHAP. Results show that (1) perceived attractiveness and visitation share a multi-core spatial structure, while visitation displays a polarized long-tail distribution; (2) attraction attributes, especially official designation, visual quality, and ticket pricing, dominate both metrics but shape them differently; (3) positively perceived social, economic, scientific, historic and aesthetical values increase visitation, whereas political value is negatively associated; and (4) accessibility and spatial agglomeration also contribute positively. Practically, multi-scale benchmarking identifies underperforming attractions, while attraction-scale value profiles and the identified influencing factors guide targeted strategies.

1. Introduction

Tourism is among the world's most dynamic economic sectors and a key driver of urban development, generating income, employment, and tax revenues while reshaping cities' spatial structures and social life (Crouch & Ritchie, 1999; Theobald, 2012; Li et al., 2018; Koufodontis & Gaki, 2022; Mor, & DalyotRam, 2023). According to the World Travel & Tourism Council (WTTC, 2025), travel and tourism contributed around US$10.9 trillion to global GDP (approximately 10%) and supported 357 million jobs worldwide. The 2030 Agenda also positions tourism as a lever for sustainable development, with Target 8.9 explicitly calling for policies that foster sustainable tourism to create jobs and promote local culture and products (United Nations, 2015).

To harness this economic potential and achieve the developmental goals, tourism planning and management must understand why tourist visitation concentrates unevenly across destinations and across attractions within destinations (Formica & Uysal, 2006). This is fundamentally a question of attractiveness, commonly understood as a place's capability to satisfy tourists' needs through a combination of attributes and experiences (Leiper, 1990; Hu & Ritchie, 1993; Crouch & Ritchie, 1999; Vengesayi et al., 2009; Reitsamer et al., 2016; Li et al., 2026). Rather than being an intrinsic property of the place alone, attractiveness emerges from interactions among (i) the attraction's attributes and values, (ii) supporting factors that facilitate access and shape on-site experience, (iii) tourists and their evaluations, and (iv) markers that shape destination image (MacCannell, 1976; Ferrario, 1979; Lew, 1987; Leiper, 1990; Crouch & Ritchie, 1999; Formica & Uysal, 2006; Vengesayi et al., 2009; Rui et al., 2025). Reflecting this multi-element nature, attractiveness research has followed two complementary traditions: a supply-side tradition that inventories attractions' physical attributes and resource endowments, and a demand-side tradition that examines how tourists perceive and evaluate those attractions (Lew,

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1987; Hu & Ritchie, 1993; Formica & Uysal, 2006; Cracolici & Nijkamp, 2009; Reitsamer et al., 2016).

In an increasingly homogenized tourism market, attractions gain competitive advantages when their unique qualities, namely what is interesting and/or unusual to see or do (Ferrario, 1979; Vengesayi et al., 2009), are positively perceived by tourists (Crouch & Ritchie, 1999; Reitsamer et al., 2016). Tourists' perceived attractiveness is therefore a critical demand-side link between attraction-side offerings and tourists' subsequent visitation behavior (Cracolici & Nijkamp, 2009). It captures tourists' subjective evaluations of an attraction's unique values and experience quality through their interaction with its attributes and supporting factors (Formica & Uysal, 2006; Giambona & Grassini, 2020; Musolino & Kotosz, 2024; Reitsamer et al., 2016). However, positive perceived attractiveness does not always translate into higher visitation, since this link may also be constrained by limited accessibility, inadequate supporting factors, and broader city-level conditions (Vengesayi et al., 2009; Reitsamer et al., 2016; Zhao et al., 2025; Li et al., 2026). Understanding tourism attractiveness and visitation therefore requires effectively measuring both sides and clarifying two linked mechanisms: how attraction attributes, supporting factors, and city-level conditions influence tourists' perceived attractiveness, and how perceived attractiveness together with these explanatory factors influences visitation.

Existing empirical work has examined parts of this chain but has not fully integrated them in large attraction systems, mainly due to measurement limitations. At regional or other large scales, studies typically operationalize tourism attractiveness through resource-based indicators of attraction endowments, tourism services and facilities, and related contextual variables (Ferrario, 1979; Formica & Uysal, 2006; Giambona & Grassini, 2020), while approximating visitation with tourist arrivals (Giambona & Grassini, 2020; Musolino & Kotosz, 2024). Tourist perceived attractiveness has also been captured through surveys, where tourists evaluate the overall quality of destination attributes and facilities, enabling comparisons of relative attractiveness across competing destinations or regions (Hu & Ritchie, 1993; Cracolici & Nijkamp, 2009, Sirkis et al., 2022). Although these approaches are effective for comparing and benchmarking destinations and identifying spatial concentration, they characterize destinations in terms of resource endowments, tourists' holistic impressions, and visitation of a destination as a whole, rather than measuring these dimensions for individual attractions within destinations. They therefore offer limited insight into how individual attractions' conditions and tourist perceptions interact to produce observed patterns. Conversely, at finer case-study scales, perceived attractiveness studies often rely on questionnaire-based measures of tourist satisfaction and attitudes, typically for one or a small number of attractions or scenes (Vengesayi et al., 2009; Reitsamer et al., 2016; Li & Wan, 2025). These studies deepen understanding of tourist-side evaluations but are difficult to scale to large attraction systems due to time and cost constraints (Zhao et al., 2024). Moreover, fine-scale indicators capturing attraction attributes, supporting factors, and city-level conditions have rarely been jointly measured and integrated to explain both perceived attractiveness and visitation.

Overall, a coherent and scalable framework remains lacking that can (i) quantify both perceived attractiveness and visitation for large numbers of attractions and aggregate them to the city scale; and (ii) systematically connect perceived attractiveness to attraction attributes, supporting factors, and city-level conditions, and examine how perceived attractiveness and these factors shape visitation.

In the Web 2.0 and platform-mediated era, social media has transformed tourists into active co-creators of attraction values by enabling them to record and share their experiences and interpretations (Ginzarly et al., 2019; Ginzarly & Srour, 2022; Ghermandi et al., 2023; Luo et al., 2025). These digital traces, commonly referred to as user-generated content (UGC), shape destination image and influence travel decision-making, functioning as contemporary markers that signal what is worth visiting and how attractions are interpreted (Du, & LiechtySantosPark, 2022; Koufodontis & Gaki, 2022; Li et al., 2026). This highlights the need to incorporate UGC into tourism attractiveness research. More importantly, UGC provides attraction-scale information on both experience and visitation: review content reflects tourists' perceptions (Koufodontis & Gaki, 2022; Zhao et al., 2024; Luo et al., 2025), while review volume serves as a proxy for visitation (Li et al., 2018; Ghermandi, 2022, Mor, & DalyotRam, 2023). Although perceptions were previously difficult to measure due to the unstructured nature of UGC, large language models (LLMs) show promise for extracting and quantifying perceived attractiveness from massive corpora (Luo et al., 2025; Rui et al., 2025; Zhao et al., 2026). Moreover, because UGC is explicitly linked to geolocated attractions, attraction-level measures of perceived attractiveness and visitation can be aggregated to the city scale, and these explicit locations also facilitate the computation of explanatory variables from spatial data.

Historic cities constitute a vital category of urban destinations where heritage attractions serve as core tourism assets. In China, these attractions are central to the identity and tourism economy of many historic cities, and their attractiveness is closely tied to how tourists perceive their multidimensional values. Focusing on 2128 heritage attractions across 143 historic cities, we propose a tourist-centered framework to quantify perceived attractiveness and visitation and to systematically connect both to explanatory variables. Perceived attractiveness is measured through an index of eight cultural values extracted from reviews using LLMs, while visitation is proxied by review volume. We quantify both metrics at the attraction, intra-city, and inter-city scales to reveal their spatial and temporal patterns and to identify attractions, cities, and time periods of relatively high or low performance. Furthermore, we compile a comprehensive set of variables capturing attraction attributes, supporting factors, and city-level conditions from multiple data sources and apply XGBoost with SHAP analysis to examine their nonlinear relationships with perceived attractiveness and visitation. Together, this framework provides diagnostic evidence for sustaining high-performing attractions and cities, improving underperforming ones, and designing targeted interventions that align heritage-value enhancement with visitation management.

Specifically, we address two research questions.

RQ1. What are the spatial and temporal patterns of tourists' perceived attractiveness and visitation across China's 143 historic cities?

RQ2. How do attraction attributes, supporting factors, and city-level conditions influence perceived attractiveness, and how do these factors and perceived attractiveness jointly shape visitation?

2. Literature review

2.1. The attraction system: theoretical foundations

Research on tourist attractiveness has long been organized around a systems perspective, whose central claim is that attractiveness is not an intrinsic property of an attraction but a relational outcome co-produced by attractions, supporting factors, tourists, and markers. Early formulations established the building blocks of this view: MacCannell (1976) conceptualized attraction as a triadic relationship among sight, tourist, and marker; Ferrario (1979) located the basic appeal of tourist resources in something interesting or unusual to see or do; and Lew (1987) linked attraction research to attraction attributes, spatial organization, and tourists' perceptions and experiences. Leiper (1990) integrated these strands into the tourist attraction system, in which attractions act as nuclei around which tourism flows develop, connected to tourists through markers and embedded in supporting spatial and functional contexts. Crouch and Ritchie (1999) extended this systems logic to destination competitiveness by distinguishing core resources and attractions from supporting factors and contextual conditions, while Vengesayi, Mavondo et al. (2009) empirically showed that attractions are the core determinants of attractiveness, with support services and people-related factors playing complementary roles.

Within this system, an attraction's unique qualities, as positively perceived by tourists, constitute the essential motivating forces for travel and the fundamental reasons why visitors choose one attraction over another (MacCannell, 1976; Ferrario, 1979; Crouch & Ritchie, 1999; Formica & Uysal, 2006; Vengesayi et al., 2009). Yet the demand side of perceived attractiveness, that is, how tourists perceive these qualities, and the supply-side attraction attributes and supporting factors that condition them are seldom measured together at the individual-attraction scale across large systems, and their joint bearing on visitation remains underexplored. Building on this foundation, our study derives perceived attractiveness by quantifying attractions' unique qualities from UGC and examines how attraction attributes, supporting factors, and city-level conditions shape these perceptions and, together with them, drive visitation. Closing this gap helps heritage managers and urban planners diagnose why individual attractions under- or over-perform within a city's portfolio, prioritize investment and interpretation across competing attractions, and design targeted interventions to strengthen the attraction system.

2.2. Perceived attractiveness of heritage attractions through cultural values

Tourist attractions encompass diverse natural and built resources (Lew, 1987; Leiper, 1990; Gunn & Var, 2002; Formica & Uysal, 2006; Vengesayi et al., 2009). In China, heritage attractions are central to many cities' tourism products and identity. Since 1982, the official list of National Famous Historical and Cultural Cities has identified cities with an exceptional concentration of cultural relics and integrated their protection into legally binding urban planning and socio-economic development strategies, as stipulated in the *Cultural Relics Protection Law of the People's Republic of China*. This policy framework positions heritage-rich attractions as strategic assets for urban development, city branding, and tourism-led regeneration.

In the World Heritage system, the unique qualities of heritage sites are formalized as Outstanding Universal Value (OUV) (UNESCO, 2017), which functions as a globally recognized standard shaping destination image and visitor expectations (Koufodontis & Gaki, 2022). Although most heritage sites do not meet the standard of OUV, several mature cultural value frameworks offer alternatives. Veldpaus (2015) synthesized frameworks from the Venice Charter, World Heritage Convention, and the Recommendation on the Historic Urban Landscape from the 1960s to 2010s, showing that they can be grouped into the eight cultural values defined by Roders (2007): social, economic, political, historic, aesthetical, scientific, age, and ecological (detailed definitions are available in Appendix A). This typology has been successfully applied to heritage monitoring and decision-making (Silva & Roders, 2012, Havinga, & ColenbranderSchellen, 2020; Foroughi et al., 2023). Compared with a single rating or an overall satisfaction score, this multidimensional account preserves which specific values tourists perceive and how strongly, we therefore adopt it as the framework through which perceived attractiveness is operationalized from tourist-generated reviews.

2.3. Measuring perceived attractiveness and visitation: UGC and LLM-based approaches

UGC offers large-scale observations, high spatiotemporal resolution, and broad geographic coverage, enabling multi-scale analyses that are difficult to achieve with conventional surveys (Coles, & DuvalHall, 2012; Li et al., 2018; Ghermandi, 2022; Bai, 2023, Mor, & DalyotRam, 2023; Zhao et al., 2024). For visitation, UGC volume indicators, including counts of geotagged photos, posts, and platform reviews, are widely used to approximate relative visitation intensity and spatiotemporal popularity (Li et al., 2018; Ghermandi, 2022, Mor, & DalyotRam, 2023). Although these indicators do not capture absolute visitor counts, meta-analytic evidence shows that they are strongly correlated with observed visits, supporting their use for relative visitation analysis (Ghermandi, 2022).

Measuring perceived attractiveness is more challenging because it concerns a latent demand-side evaluation. Conceptually, it belongs to the broader domain of tourist perception, understood here as a post-visit appraisal in which attraction attributes are interpreted through tourists' cognitive judgments, affective responses, and prior tourism-related knowledge (He, & LiuSongQiuZhang, 2026; Hu & Ritchie, 1993; Reitsamer et al., 2016; Yang & Hsu, 2026). This view connects perceived attractiveness to several post-experience constructs widely used in tourism research. These include satisfaction, which cognitive-affective theory frames as an outcome shaped by cognitive evaluations of service or experience attributes and affective responses to them (He, & LiuSongQiuZhang, 2026); attitude and attachment, which capture tourists' evaluative responses and place-related bonds formed through experience (Reitsamer et al., 2016); and sentiment, which may differ across aspects of the same attraction, generating positive, negative, or ambivalent evaluations simultaneously (Yang & Hsu, 2026). Such evaluations are not directly observable, but after a visit tourists externalize them in review narratives, which disclose what attributes and values they attend to and whether they appraise them positively or negatively (He, & LiuSongQiuZhang, 2026; Yang & Hsu, 2026). Online reviews are therefore a theoretically appropriate basis for measuring perceived attractiveness.

Measuring perceived attractiveness, however, requires methods that preserve this multidimensional structure rather than flattening it into a single score. Platform ratings offer a simple overall indicator of tourist evaluation (Huang, & ShiLiWang, 2025; Li et al., 2026), yet they compress heterogeneous perceptions into one score and often show limited variance, reducing their ability to discriminate among attractions. Tourism text-analysis research has widely transformed online reviews into metrics of tourist satisfaction and sentiment, most commonly through sentiment analysis using lexicon-, machine-learning-, and deep-learning-based methods (Bi, & ZhuHan, 2024). However, conventional sentiment analysis usually collapses multidimensional narratives into a single polarity score and therefore struggles with domain-specific, multidimensional cultural-value judgments and reviews that mix positive and negative aspects (Zhao et al., 2024; Rui et al., 2025; Yang & Hsu, 2026). Domain-adapted language models partly address this limitation by linking sentiment to explicit experience dimensions. For example, Yang and Hsu (2026) conceptualized tourist sentiment ambivalence as arising from aspect-level sentiments and incorporated tourism domain knowledge into BERT to classify sentiments toward attraction aspects such as price, amenities, transport, commercialization, and scenery. More directly related to cultural value measurement, Foroughi, de Andrade et al. (2023) used a BERT-based classifier to categorize online posts about the windcatchers of Yazd into cultural-value categories, showing that public perceptions of heritage values can be extracted from UGC. However, these supervised or domain-adapted approaches usually depend on annotated samples, domain lexicons, or additional model training, which constrains their transferability to large and heterogeneous UGC corpora where expressions are noisy, context-dependent, and unevenly distributed across attractions and value dimensions.

Recent LLM-based approaches provide a promising route for operationalizing multidimensional perception measurement. Compared with conventional text-mining pipelines, LLMs can translate theoretical definitions, coding criteria, and illustrative examples into prompts, enabling unstructured UGC to be coded into dimension-specific labels and scores with little or no task-specific training data. In tourism research, Sun, Yin et al. (2026) proposed a human-AI collaborative workflow that uses zero-shot prompting to extract tourist perception features from Tripadvisor reviews, refine them into a hierarchical evaluation system, and assign sentiment scores to specific perception dimensions. Luo, Zhang et al. (2025) showed that few-shot and definition-based prompts can identify cultural ecosystem service

categories and associated sentiments from reviews without task-specific model training. Synthesizing these studies, we frame LLM-based perception measurement as a conceptual model that applies three operations to the externalization logic established above: theory pre-defines the perceptual dimensions to be recovered (here, the eight cultural values); prompts encode these definitions as semantic-coding criteria and illustrative examples; and the model returns review-level dimension labels and sentiment scores that aggregate into attraction-level perception metrics. Nevertheless, whether LLMs can reliably extract the eight cultural values as a measure of perceived attractiveness, and whether the resulting value-based perceptions explain visitation across large attraction systems, remains insufficiently examined.

2.4. Underlying mechanisms: explanatory variables and nonlinear relationships

The attraction system theories imply that perceived attractiveness and visitation arise not from any single factor but from how attraction attributes, supporting factors, and broader city context act together (Leiper, 1990; Crouch & Ritchie, 1999; Vengesayi et al., 2009). Their effects are therefore likely to be interdependent and to vary in strength across different ranges rather than being constant and additive, as increasingly evidenced by empirical studies of tourism and urban perception (Rui et al., 2025; Zhao et al., 2025; Li et al., 2026). Capturing such mechanisms therefore calls for measuring fine-scale conditions at multiple levels and for models able to represent nonlinear, threshold-like relationships, neither of which has been fully realized in attractiveness and visitation research across large attraction systems. Related work in urban studies offers mature approaches on both fronts. For measurement, supporting factors can be quantified through established built-environment metrics, most notably the 5D framework of Density, Design, Diversity, Distance to transit, and Destination accessibility, which has been widely applied to capture urban spatial and functional structures (Huang, & ShiLiWang, 2025; Rui et al., 2025). For modeling, combining such multidimensional metrics with machine-learning models such as XGBoost and explainability tools such as Shapley additive explanations (SHAP) enables the identification of nonlinearities and thresholds that the systems view anticipates but that linear models obscure (Huang, & ShiLiWang, 2025; Rui et al., 2025; Zhao et al., 2025).

3. Materials and methods

3.1. Theoretical framework and research workflow

Our theoretical framework (Fig. 1) conceptualizes perceived attractiveness and visitation as outcomes of the interaction between tourists and attractions within a multi-scale urban context. Attraction attributes, supporting factors, and city-level conditions jointly influence perceived attractiveness, which in turn, together with these factors, shapes visitation. In this study, attraction attributes capture the core characteristics of the attractions themselves, whereas supporting factors describe the external accessibility and conditions that facilitate tourists' access to and experience of an attraction.

Fig. 2 illustrates the research workflow. Step 1 compiles UGC and multi-source data for 2128 attractions across 143 cities. Step 2 quantifies perceived attractiveness and visitation from UGC using LLMs. Step 3 examines spatial and temporal patterns across attraction, intra-city, and inter-city scales to identify attractions, cities, and periods with relatively high or low performance. Step 4 links both perceived attractiveness and visitation to the explanatory variables. By integrating the diagnostic patterns from Step 3 with the explanatory insights from Step 4, we identify where underperformance arises from experience-quality deficits or limited visitation, and derive targeted strategies to sustain high performance and improve low performance.

3.2. Quantifying and exploring spatial and temporal patterns of perceived attractiveness and visitation

3.2.1. UGC data collection and attraction selection

The study area covers the 143 National Famous Historical and Cultural Cities officially designated by the end of 2025. Using Ctrip, a leading Chinese travel platform widely adopted in urban and tourism research (Zhao et al., 2024; Luo et al., 2025; Rui et al., 2025), we identified heritage attractions with more than 50 reviews, yielding 2128 attractions and 619,429 review records. After deduplication and cleaning, the final dataset comprised 610,136 valid reviews and 940,477 images. Fig. 3 maps the 2128 attractions, with dense clusters around Beijing, Shanghai, Chengdu, and Guangzhou.

3.2.2. Quantifying perceived attractiveness and visitation from review texts

Perceived attractiveness is operationalized from reviews using an

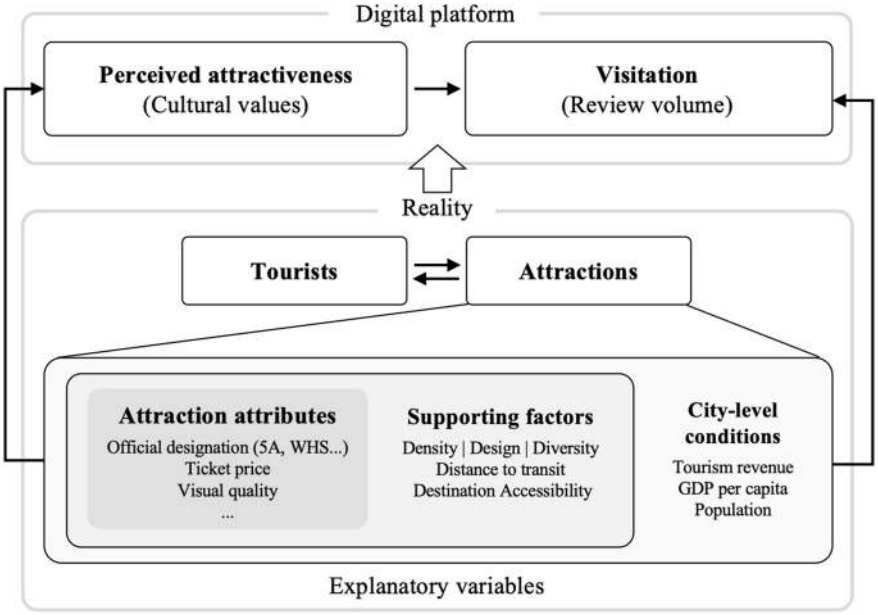


Fig. 1. Theoretical framework.

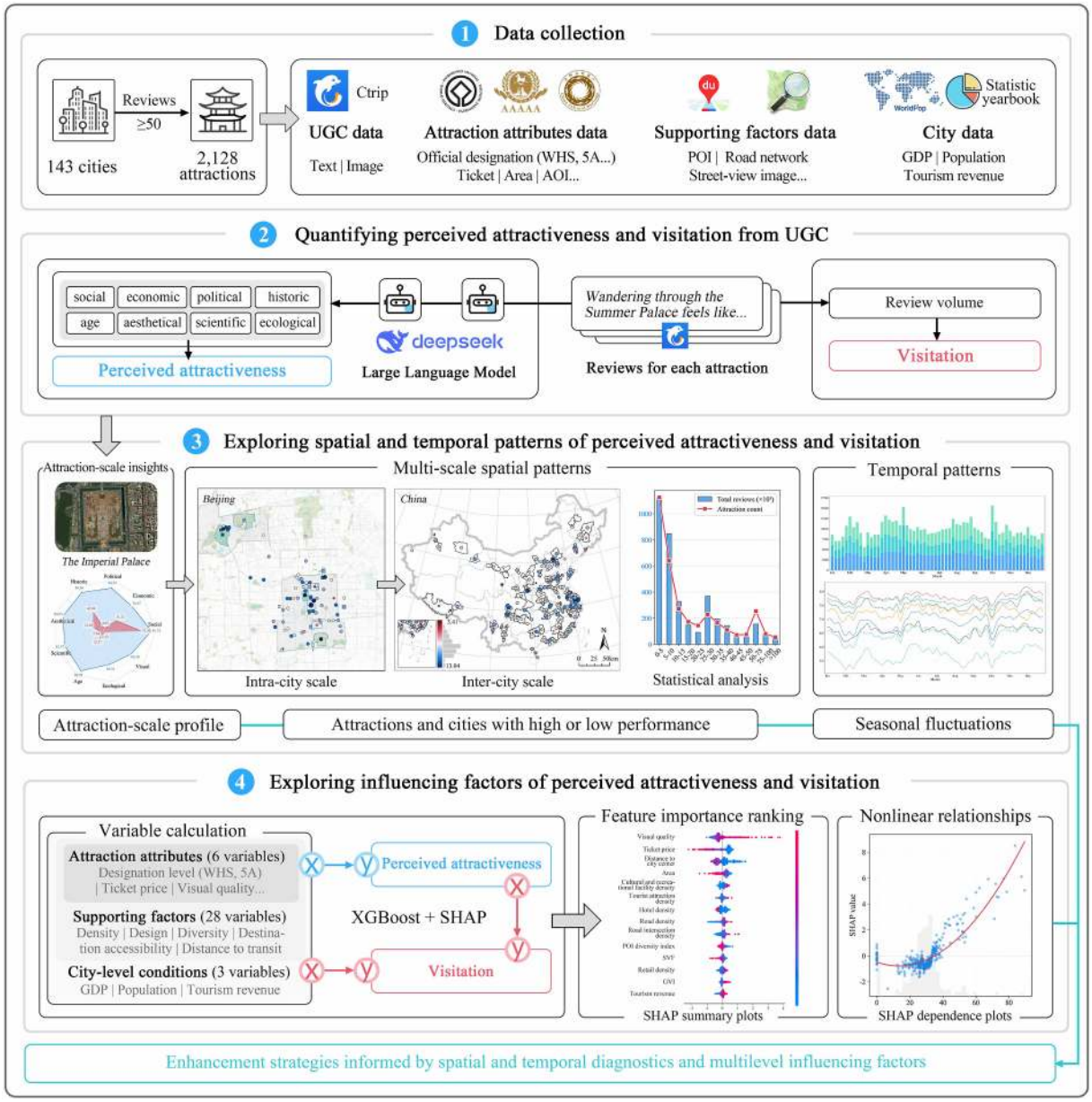


Fig. 2. Research workflow.

index of eight cultural values: social, economic, political, historic, aesthetical, scientific, age, and ecological (definitions in Appendix A) (Roders, 2007; Silva & Roders, 2012; Havinga, & ColenbranderSchellen, 2020; Foroughi et al., 2023). Fig. 4 illustrates the LLM-based pipeline that quantifies reviews to cultural-value scores, which proceeds as follows.

(i) **Input and contextual calibration.** Three pre-defined examples are provided for few-shot prompting. Each review text x_t is additionally processed with a rolling-window in context prompt memory M_t, which appends the three most recent successful exemplars as additional few-shot demonstrations:

$$M_t = \{(x_{t-k}, y_{t-k})\}_{k=1}^{3} \tag{1}$$

where x_{t-k} is the k-th most recent review text whose extraction succeeded, and y_{t-k} is its verified structured output.

(ii) **Preliminary extraction.** An LLM agent extracts candidate, dimension-wise evidence from x_t based on the eight cultural-value definitions (using Chain-of-Thought reasoning):

$$\widetilde{\mathbf{E}}_t = f_\theta(x_t; M_t) = \left[\widetilde{E}_{t,d}\right]_{d=1}^{8} \tag{2}$$

where $\widetilde{E}_{t,d}$ is the set of sentences extracted as traceable evidence supporting dimension d.

(iii) **Verification and scoring.** A second agent executes a two-stage process. First, it removes any extracted evidence not appearing verbatim in the original review (hallucination filtering) and then removes evidence inconsistent with the dimension definition (relevance validation), yielding verified evidence $E_{t,d}$. Second, based on $E_{t,d}$, it assigns a dimension-level perception score $s_{t,d}$ within the range [1, 10]; if no valid evidence remains, the dimension is marked N/A:

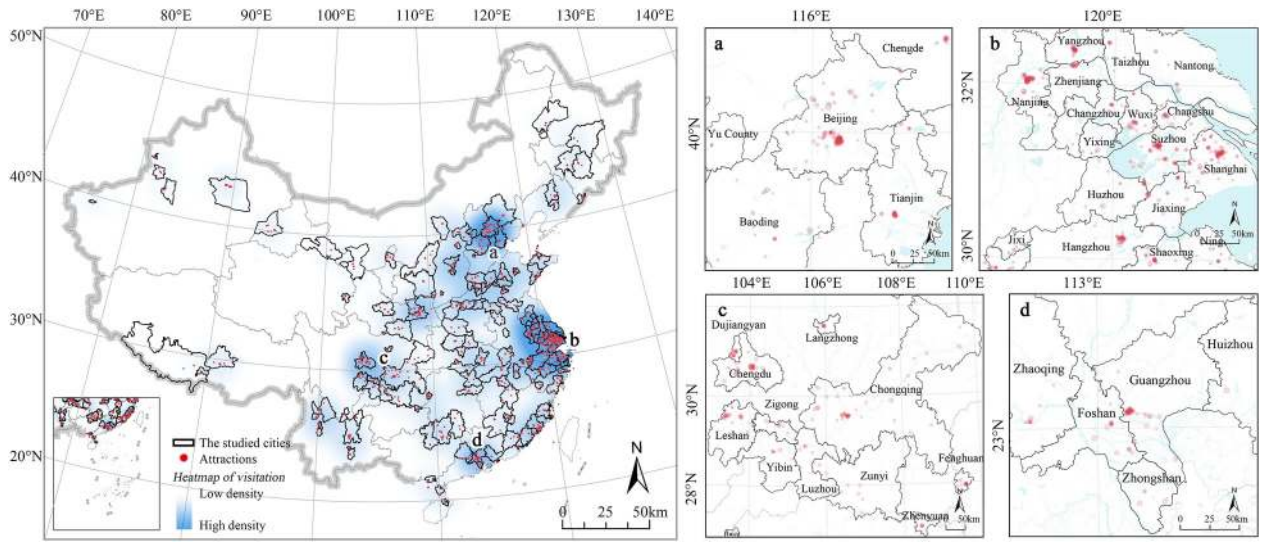


Fig. 3. Spatial distribution of the 2128 studied attractions across 143 cities. Panels (a-d) highlight major high-density clusters, where red points denote attractions and point size reflects visitation.

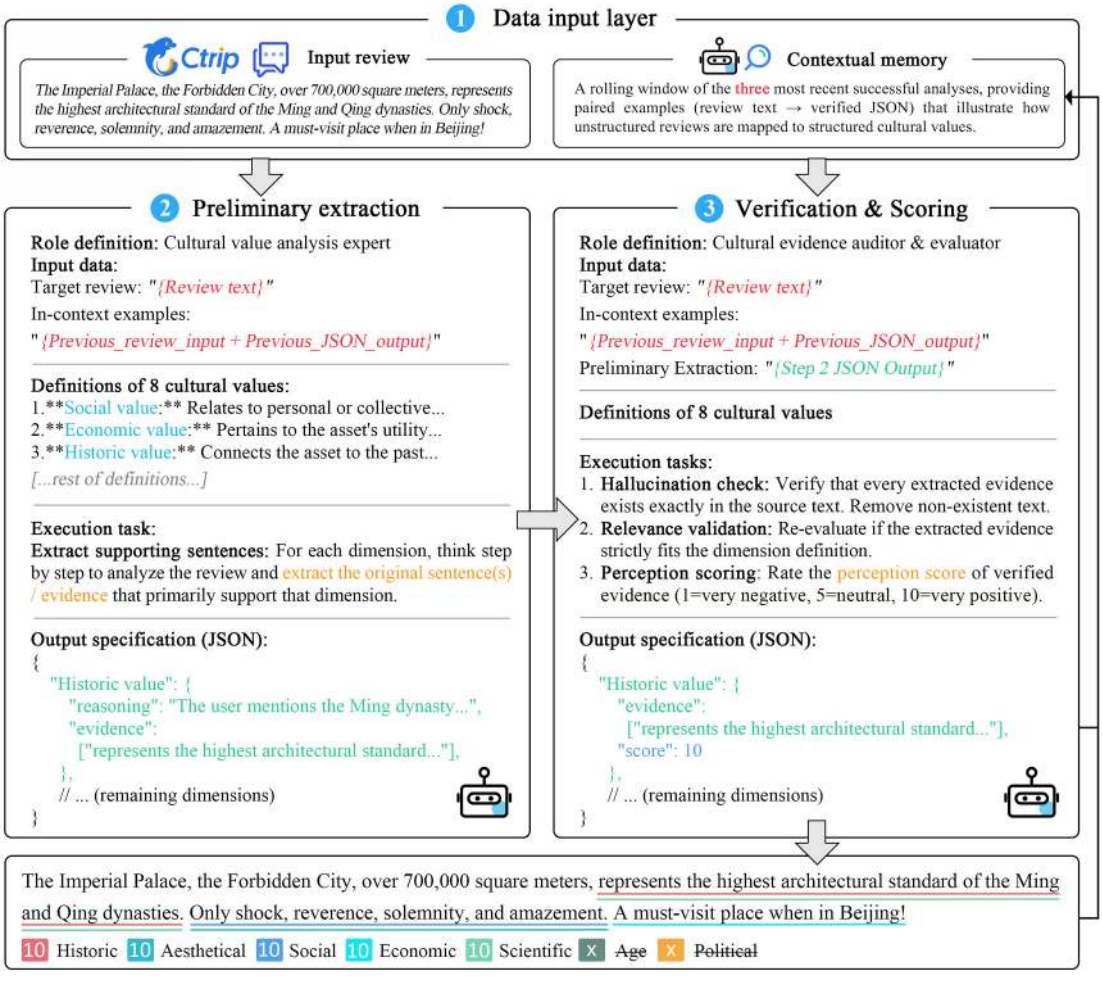


Fig. 4. LLM-based pipeline of quantifying eight cultural-value scores from reviews.

$$s_{t,d} = \begin{cases} h_d(E_{t,d}), \text{if } E_{t,d} \neq \varnothing \\ \text{N/A}, \text{if } E_{t,d} = \varnothing \end{cases} \quad (3)$$

where $h_d(\cdot)$ denotes the function mapping verified evidence to a perception score.

After quantifying eight cultural-value scores for each review, we calculate each attraction's perceived attractiveness by aggregating these scores. To stabilize estimates for attractions with sparse reviews, we

apply Bayesian smoothing to calculate smoothed score $P_{a,d}$ for each dimension:

$$P_{a,d} = \frac{N_{a,d}R_{a,d} + C_d m_d}{N_{a,d} + C_d} \quad (4)$$

where $N_{a,d}$ is the number of reviews for attraction a containing valid evidence for dimension d, $R_{a,d}$ is the corresponding mean score, m_d is the global mean score for dimension d across all attractions, and C_d is a confidence parameter defined as the mean number of reviews per attraction for that dimension.

We min-max normalize $P_{a,d}$ within each dimension to obtain $P'_{a,d}$. Overall perceived attractiveness for attraction a is then computed as a frequency-weighted aggregation:

$$P_a = \frac{\sum_{d=1}^{8} P'_{a,d} N_{a,d}}{\sum_{d=1}^{8} N_{a,d}} \quad (5)$$

A city's perceived attractiveness is obtained by aggregating P_a across all its attractions.

We benchmarked six LLMs (including DeepSeek-V3.2-Exp, GPT-5-mini, Gemini-2.5-Flash, Doubao-Seed-1.6, Qwen3-30B-A3B-Instruct-2507, and GLM-4.5) against a manually annotated dataset of 1500 reviews, utilizing tolerance-1 accuracy (denoted as $\mathrm{Acc}_{\leq 1}$) to select the optimal model.

$$\mathrm{Acc}_{\leq 1} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(|y_i - \hat{y}_i| \leq 1) \quad (6)$$

where N is the total number of samples, $y_i \in \{1, \ldots, 10\}$ is the ground-truth rating, $\hat{y}_i \in \{1, \ldots, 10\}$ is the predicted rating, and $\mathbb{I}(\cdot)$ is the indicator function.

Visitation volume for an individual attraction is proxied by review volume. City-level visitation volume is calculated by aggregating review volumes across all constituent attractions.

3.2.3. Spatial and temporal analysis

At the attraction and intra-city scales, we selected representative attraction-rich cities to present spatial patterns of quantified attractiveness. Areas of Interest (AOIs) were retrieved from Baidu Map (https://lbsyun.baidu.com/), and all coordinates were manually verified and corrected. At the inter-city scale, we mapped perceived attractiveness and visitation and the eight cultural values, and applied Global and Local Moran's I in ArcGIS Pro to identify spatial clusters. For statistical comparisons and temporal analyses, cities were grouped into population-based tiers: Tier 1-2 (>5 million), Tier 3-4 (1-5 million), and Tier 5-6 (<1 million).

3.3. Exploring relationships between perceived attractiveness and visitation and explanatory variables

We incorporated multi-source datasets (detailed in Appendix B) to construct explanatory variables (see Table 1). City-level conditions include tourism revenue, population, and GDP per capita. The attraction attributes and supporting factors are described below.

3.3.1. Attraction attributes

(1) *Official designation*

We compiled three authoritative inventories, including UNESCO World Heritage Sites (WHS), National Key Cultural Relics Protection Units (NKCRPU), and A-Level Tourist Attractions (5A, 4A, and 3A). Among them, WHS properties are internationally designated sites recognized for their Outstanding Universal Value. NKCRPU sites are designated by the State Council of China as the highest tier of nationally protected immovable cultural relics, capturing national heritage significance and conservation priority. A-Level Tourist Attractions are certified by China's Ministry of Culture and Tourism on a 1-5A scale, with 5A representing the highest standard of tourism resources, visitor facilities, and management quality. Considering attraction names and recorded locations differ across sources, we employed a multi-stage matching framework (Fig. 5a) to align designations to the attractions.

For NKCRPU and A-Level lists, we applied exact name matching and then semantic matching using Seed-1.6-Embedding vectors and cosine similarity:

$$S_c(A, B) = \frac{A \cdot B}{\|A\|\|B\|} \quad (7)$$

The highest similarity candidates were further screened by DeepSeek-V3.2-Exp to judge whether paired names refer to the same attraction, followed by manual verification.

For WHSs, direct name matching is often insufficient because WHS properties often encompass multiple components or serial sites. We therefore adopted an LLM-augmented web retrieval approach using Doubao-Seed-1.6 to determine whether the attraction is a listed property or a component of a listed property, followed by manual verification. Remaining ambiguities were resolved by spatial matching in ArcGIS Pro, checking whether an attraction's AOI overlapped the designated

Table 1
Descriptive statistics of variables (N = 2128).

Variable	Mean	SD.	Min.	Max.
Attraction attributes				
NKCRPU score	2.09	2.94	0.00	37.00
A-level score	3.99	0.67	3.00	5.00
World Heritage Status	0 (86.00%), 1 (0.80%), 2 (13.20%)			
Visual quality	31.07	11.40	0.00	100.00
Ticket price (CNY)	17.16	29.81	0.00	238.00
Area (ha)	35.09	95.53	0.02	809.90
Supporting factors				
Density (within a 500 m buffer; count/km^2)				
Total POIs	708.60	811.33	0.00	6158.54
Restaurants	135.60	168.33	0.00	1301.28
Hotels	50.57	122.79	0.00	1348.86
Retail	227.08	306.97	0.00	1585.21
Tourist attractions	23.56	39.40	0.00	318.75
Cultural and recreational facilities	3.28	5.56	0.00	49.77
Service facilities	87.33	101.38	0.00	763.12
Bus stops	5.51	5.22	0.00	32.08
Subway stations	0.26	0.61	0.00	6.50
Population density	8078.78	7592.61	0.00	36553.47
Design (within a 500 m buffer)				
Road intersection density (count/km^2)	369.17	442.69	0.21	3961.47
Road density (km/km^2)	10.56	7.19	0.71	44.16
Green View Index (GVI)	0.22	0.11	0.00	0.80
Sky View Factor (SVF)	0.17	0.08	0.00	0.47
Enclosure	0.44	0.10	0.05	0.83
Walkability	0.05	0.02	0.00	0.19
Beautiful	5.66	0.48	3.89	7.51
Wealthy	5.68	0.36	4.09	7.24
Safe	5.83	0.43	3.50	7.91
Lively	5.98	1.17	2.71	7.98
Depressing	5.82	0.37	4.27	7.61
Boring	6.49	0.33	4.99	7.74
Diversity (within a 500 m buffer)				
Functional mixture	3.49	0.86	0.00	4.73
Distance to transit (km)				
Distance to nearest subway station	13.68	17.40	0.29	233.29
Distance to nearest bus station	2.66	12.37	0.01	229.53
Destination accessibility (km)				
Distance to nearest train station	14.09	16.87	0.07	121.91
Distance to nearest airport	38.59	26.42	2.45	162.32
Distance to city center	19.40	25.21	0.03	263.52
City-level conditions				
Population (million persons)	4.64	5.19	0.07	31.90
GDP per capita (thousand CNY per person)	91.40	47.60	24.20	228.00
Tourism revenue (billion CNY)	94.25	95.90	0.13	672.24

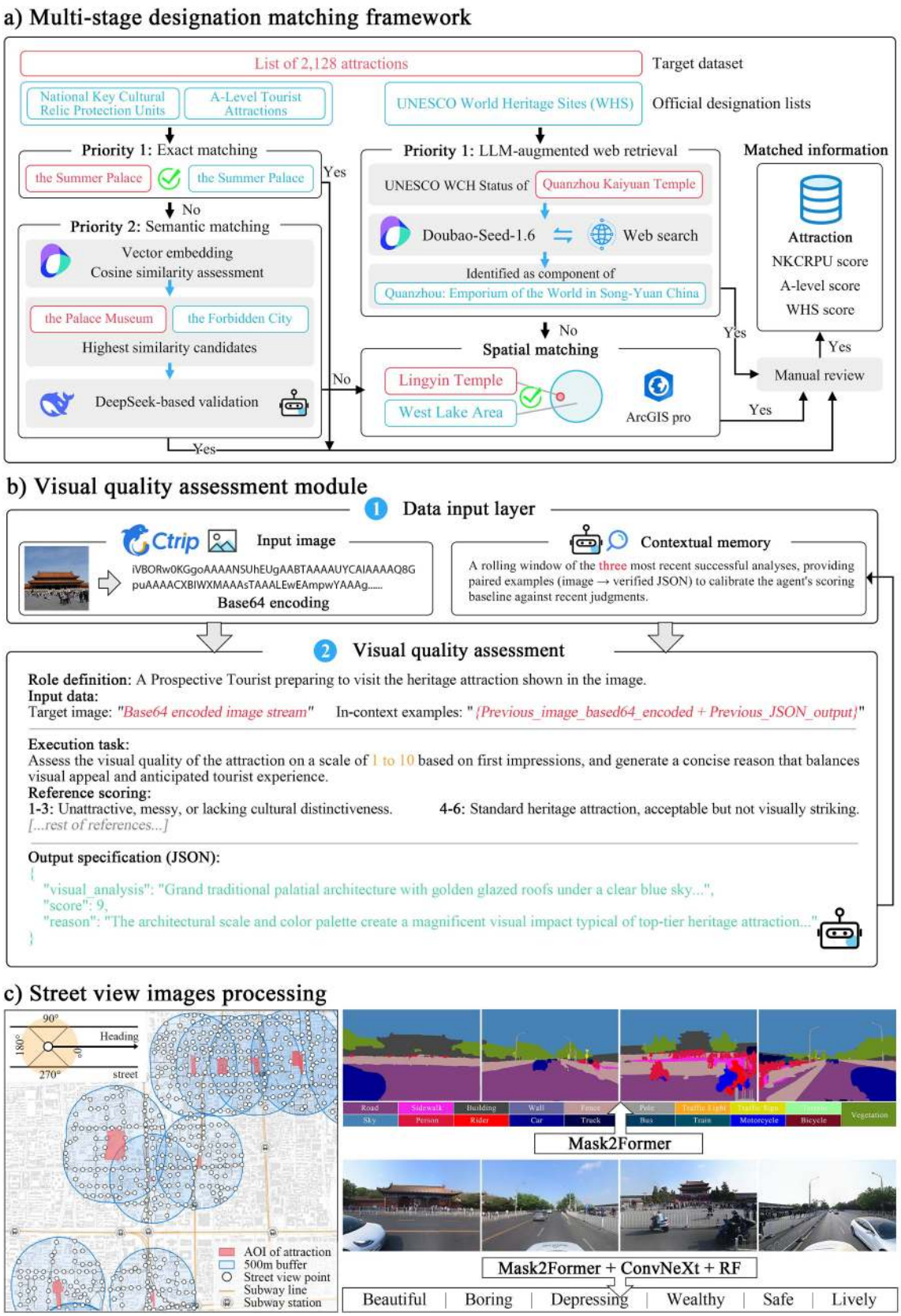


Fig. 5. Major methods for variable calculation. (a) Multi-stage framework for linking attractions with official designations. (b) VLM-based evaluation of review image visual quality. (c) SVI processing procedure.

attraction zone identified in earlier steps.

We constructed three designation variables from the matching results. The NKCRPU score applies inverse weights to the eight official batches (8 = earliest, 1 = latest) and aggregates weights for attractions containing multiple NKCRPUs. The A-level score is coded as 5, 4, or 3 for 5A, 4A, and 3A sites. World Heritage Status is coded as 1 (designated site), 2 (within a WHS), or 0 (non-designated). Unmatched attractions are also coded 0.

(2) *Visual quality of attractions derived from review images*

Unlike review texts, which directly represent tourist opinions, review images primarily depict on-site attraction scenes, functioning similarly to street view images (SVIs). We therefore treat the visual quality derived from these images as an attraction attribute. A recent GPT-4-based study has shown that image-based prompting can approximate human visual-perception assessments from urban images when validated against human ratings, supporting the use of VLMs for scalable visual quality evaluation (Malekzadeh et al., 2025). Fig. 5b shows the VLM-based visual-quality module. Each image is Base64-encoded and scored by an agent instructed to act as a prospective tourist, rating visual quality on a 1-10 scale. To improve consistency, we use the same rolling-window memory mechanism as in Fig. 4 (three most recent successful cases). The model outputs a structured JSON containing a visual description, a numerical score, and a brief justification.

To benchmark VLM scores against human perception, we created a ground-truth dataset using paired comparisons, which are generally more reliable than absolute ratings for perceived attributes (Dubey, & NaikParikhRaskarHidalgo, 2016). A total of 400 images covering diverse attraction types were sampled, and 40 students from the School of Architecture were recruited as evaluators, with a balanced gender ratio (female: male = 1:1). All participants had received formal training in architectural design, providing them with relevant expertise for assessing visual and spatial qualities in the image-rating task. The participants were aged 22-30 years (mean = 25.75, SD = 2.19), including 26 master's students (65.0%) and 14 doctoral students (35.0%). We implemented an online paired-comparison system based on the TrueSkill algorithm (details in Appendix C). In each comparison, participants were shown two images and asked which scene appeared more attractive to visit. After the experiment, TrueSkill scores were rescaled to a 1 to 10 range using min-max normalization.

We evaluated six VLMs, including GPT-4o, Gemini-2.5-Flash, Doubao-Seed-1.6-vision, Qwen3-VL-30B-A3B-Instruct, GLM-4.5V, and GLM-4V-Plus-0111, using $Acc_{\leq 1}$ against the human benchmark.

3.3.2. Supporting factors

We adopted the 5D framework comprising Density, Design, Diversity, Distance to transit, and Destination accessibility to measure supporting factors. Density, Design, and Diversity were measured within a 500 m buffer around each attraction, a commonly adopted spatial scale in urban studies (Huang, & ShiLiWang, 2025; Zhao et al., 2025). Distance to transit and Destination accessibility were calculated as distances from each attraction to the corresponding transport facilities and urban destinations.

Density variables included population density and POI densities (count per buffer area). The latter were derived from 16,290,675 POIs covering both the aggregate total and key categories for tourism and daily services, including restaurants, hotels, retail, tourist attractions, cultural and recreational facilities, service facilities, bus stops, and subway stations.

Design-related variables capture street network characteristics and the visual experience provided to tourists (Rui, Xu et al., 2025). We measured road intersection density and road density using road network data. To quantify the streetscape characteristics, we collected 339,056 SVIs at 50 m intervals within buffers. These SVIs were processed using the Mask2Former model (Cheng et al., 2022) with a Swin-L (in22k) backbone, which achieved an mIoU of 83.65 on the Cityscapes dataset, to segment 19 street elements (e.g., road, sidewalk, sky, vegetation). Following established methods (Huang, & ShiLiWang, 2025; Rui et al., 2025; Zhao et al., 2025), we calculated the Green View Index (GVI) (vegetation proportion), Sky View Factor (SVF) (sky proportion), enclosure (vertical element proportion), and walkability (an approximation of pedestrian-friendliness) from the segmented elements (details in Appendix D). Additionally, we applied a "Mask2Former + ConvNeXt + Random Forest (RF)" model trained on the Place Pulse 2.0 dataset to predict six perceptual attributes (beautiful, boring, depressing, lively, safe, and wealthy) (Zhao et al., 2025). Fig. 5c illustrates the SVI processing procedure.

Diversity is measured by the functional mixture within each buffer using the Shannon Entropy Index based on the distribution of POI categories:

$$H = -\sum_{i=1}^{n} P_i \ln P_i \tag{8}$$

where n is the number of POI categories and P_i is the proportion of category i. Higher values indicate a more even functional composition and greater land-use mixing.

Distance to transit represents local network access, measured as the distance to the nearest subway station and bus station.

Destination accessibility captures regional centrality and external connectivity, measured as the distance to the nearest train station, airport, and city government (used as a proxy for the city center).

3.3.3. Exploring relationships between explanatory variables and perceived attractiveness and visitation

Using the full sample of 2128 attractions, we conducted two predictive analyses: (i) modeling how explanatory variables influence perceived attractiveness, and (ii) modeling how these variables, together with eight perceived cultural values, collectively influence visitation. To mitigate multicollinearity, we removed predictors with Variance Inflation Factors (VIF) exceeding 10.0 using the Python statsmodels library. We partitioned the dataset into training (60%), validation (20%), and testing (20%) subsets and applied five-fold cross-validation. We benchmarked six models, including XGBoost, Gradient Boosting, Random Forest, Support Vector Regression, Ridge, and Lasso. Performance was evaluated using R^2, RMSE, and MAE.

The best-performing model was interpreted using SHAP summary and dependence plots to examine predictor importance and nonlinear relationships. SHAP provides a unified framework for interpreting any machine learning model by computing each predictor's marginal contribution to individual predictions, derived from game-theoretic Shapley values (Lundberg et al., 2020). SHAP summary plots rank predictors by their overall importance, while dependence plots reveal how the effect of a given predictor varies across its value range, enabling the identification of nonlinearities and thresholds. Detailed model settings and technical procedures are provided in Appendix E.

4. Results

4.1. Performance of LLMs and VLMs

Based on the benchmarking results in Table 2, we selected DeepSeek-V3.2-Exp for text analysis and Doubao-Seed-1.6-vision for image processing. These models provided the optimal trade-off, achieving the highest tolerance-1 accuracies of 0.862 and 0.818, respectively, while ensuring cost-effectiveness ($0.432 and $1.151 per 1M tokens) and practical runtimes (9.108s per review and 4.468s per image). The full inference task consumed roughly 721 million text tokens and 1.06 billion image tokens, and executed about two weeks using six parallel threads. Three methodological notes are relevant here. First, processing the full dataset (610,136 review texts and 940,477 images) required

Table 2
Performance, speed, and cost comparison of candidate LLMs and VLMs. The best-performing models for both tasks are highlighted.

Model	Release date	Price (Output/ 1M token)	Avg. tokens per request	Avg. processing time (s/ request)	$Acc_{\leq 1}$
Quantifying perceived attractiveness (averaging the eight cultural values) from texts					
DeepSeek-V3.2-Exp	2025/9/30	$0.432	1553.406	**9.108**	**0.862**
Doubao-Seed-1.6	2025/10/15	$0.288	1913.340	14.862	0.852
GPT-5-mini	2025/8/7	$2.000	4308.096	19.590	0.846
GLM-4.5	2025/7/30	$2.014	2255.652	18.126	0.830
Qwen3-30B-A3B-Instruct	2025/8/1	$0.575	1754.664	9.342	0.822
Gemini-2.5-Flash	2025/6/17	$2.500	5695.554	21.846	0.808
Quantifying visual quality from images					
Doubao-Seed-1.6-vision	2025/8/15	$1.151	1127.435	4.468	**0.818**
GLM-4.5V	2025/8/12	$0.863	1012.161	3.660	0.810
Qwen3-VL-30B-A3B-Instruct	2025/10/4	$0.617	751.280	**0.880**	0.793
GPT-5-mini	2025/8/7	$2.000	1052.176	6.189	0.783
Gemini-2.5-Flash	2025/6/17	$2.500	2453.056	11.248	0.768
GLM-4V-Plus	2025/1/11	$0.575	918.287	2.472	0.745

Note: Processing times were measured via direct API calls and are affected by network conditions, server load, and request throttling. They are therefore intended for practical comparison under our deployment setting in China rather than as absolute model inference speed.

balancing predictive performance with cost; we therefore excluded flagship models like Gemini-3-Pro and GPT-5.2 due to prohibitive pricing ($12.00 and $14.00/1M output tokens, respectively). Second, the average text processing time on the full dataset (5.667s) was faster than the benchmark due to shorter average review lengths. Third, for the visual-quality benchmark, our paired-comparison experiment involved an average of 15.87 comparisons per image (6348 total comparisons).

4.2. Attraction-scale insights

Fig. 6 illustrates examples of quantified cultural values derived from reviews. For each attraction, our framework provides value-specific scores and corresponding traceable textual evidence, thereby ensuring transparency and interpretability. Taking the Imperial Palace (the Palace Museum in Beijing) as an example, social value is the most frequently mentioned dimension, appearing in 72.30% of reviews, and it is also rated highly, with an average score of 91.74 out of 100. Review narratives describe strong emotional responses such as "shock, reverence, solemnity, and amazement," corresponding to the *"emotional"* subtype of social value (Appendix A ; Roders, 2007). Beyond social value, it ranks among the highest across all attractions in historic (86.36), age (89.29), aesthetical (84.67), scientific (81.37), and political (84.52) values. These scores align with review content emphasizing imperial dynasties, aesthetic qualities, craftsmanship, and political authority. In contrast, ecological value is relatively weak, being mentioned in only 2.33% of reviews and receiving an average score of 65.51. Overall, the perceived attractiveness calculated from these cultural values is not a single sentiment score, but a structured summary of what types of value tourists perceive and talk about, plus what they criticize.

4.3. Spatial and temporal patterns of perceived attractiveness and visitation

4.3.1. Intra-city-scale spatial patterns

Fig. 7a maps intra-city-scale perceived attractiveness for representative cities that contain a relatively high concentration of attractions with more than 50 reviews. Cities not shown generally have fewer attractions and more dispersed distributions. These maps distinguish well-perceived from poorly-perceived attractions, providing a direct diagnostic tool for identifying low-scoring attractions that may require targeted upgrading, which can be guided by value-specific textual evidence, as illustrated in Fig. 6.

Spatially, these attractions tend to cluster within government-designated protected historic districts, forming dense hotspots in well-established tourism destinations such as Beijing, Shanghai, Suzhou, Nanjing, Xi'an, and Guangzhou, and more spatially dispersed but still district-centered patterns in cities such as Tianjin, Qingdao, Hangzhou, and Wuhan. Attractions inside protected historic districts also demonstrate slightly higher perceived attractiveness (mean score = 59.43) and visitation (mean volume = 2060) than those outside these districts (mean score = 57.35; mean volume = 1526).

Fig. 7b quantifies a strong distance-to-city-center gradient: both attraction counts and visitation volumes decline with increasing distance from city centers. Secondary peaks emerge at intermediate or outer distances in some tiers, suggesting destination-type attractions that attract substantial visitation despite their peripheral location, such as the Great Wall in Beijing, the Terracotta Warriors in Xi'an, and the Longmen Grottoes in Luoyang. Additionally, visitation is highly polarized, as the top five attractions of each city collectively capture 85.2% of all reviews.

4.3.2. Inter-city-scale spatial patterns

Nationally, visitation also exhibits strong polarization. The top five cities, including Beijing, Xi'an, Shanghai, Suzhou, and Nanjing, account for 38.7% of all reviews, with the top ten cities contributing 52.7%. Additionally, both perceived attractiveness and visitation, as well as the eight cultural values, exhibit positive spatial autocorrelation (Global Moran's I from 0.0687 to 0.1449; $p < 0.001$; *Appendix G*, Table S4). Local indicators of spatial association (LISA) maps further reveal a multi-core structure, with consistent high-high clusters around Beijing in North China, Shanghai in East China, and Chengdu in Southwest China. Smaller, dimension-specific clusters and spatial outliers are also observed around Guangzhou, Xi'an, Quanzhou, Taiyuan, Chongqing, Wuhan, and Changsha, among others (Fig. 8 andFig. 9).

Supplementary stratification analyses (Appendix F, Fig. S4) further show that perceived attractiveness and cultural values increase near-monotonically with city tier, and that among the designation batches of National Famous Historical and Cultural Cities, Batch 1 cities rank first across all ten dimensions, Batch 2 cities rank second across most, with Batches 3 and 4 showing minimal differentiation from each other.

Fig. 9 further decomposes perceived attractiveness into eight cultural values, and Table 3 summarizes the top keywords derived from the 610,136-review corpus by aggregating the verified evidence ($E_{t,d}$) for each value and applying word segmentation and frequency analysis. Social (52.29%), aesthetical (52.22%), and economic (43.97%) values dominate online narratives. Social value is expressed through general appraisals such as "worth visiting" (30,507). The *"individual emotional"* subtype of social value (Appendix A) is suggested by terms such as "experience" (15,501), whereas the *"collective emotional"* subtype appears in family- and learning-oriented expressions such as "children" (15,240) and "learning" (11,785). Highly-rated attractions include The Palace Museum (Beijing) and West Lake (Hangzhou). Aesthetical value is anchored by visual descriptors including "beautiful" (42,155) and "scenery" (37,187), with high scores at attractions such as Shuanglin Temple (Pingyao) and the Yungang Grottoes (Datong). Economic value emphasizes tourism-oriented consumption, reflected in "worth visiting"

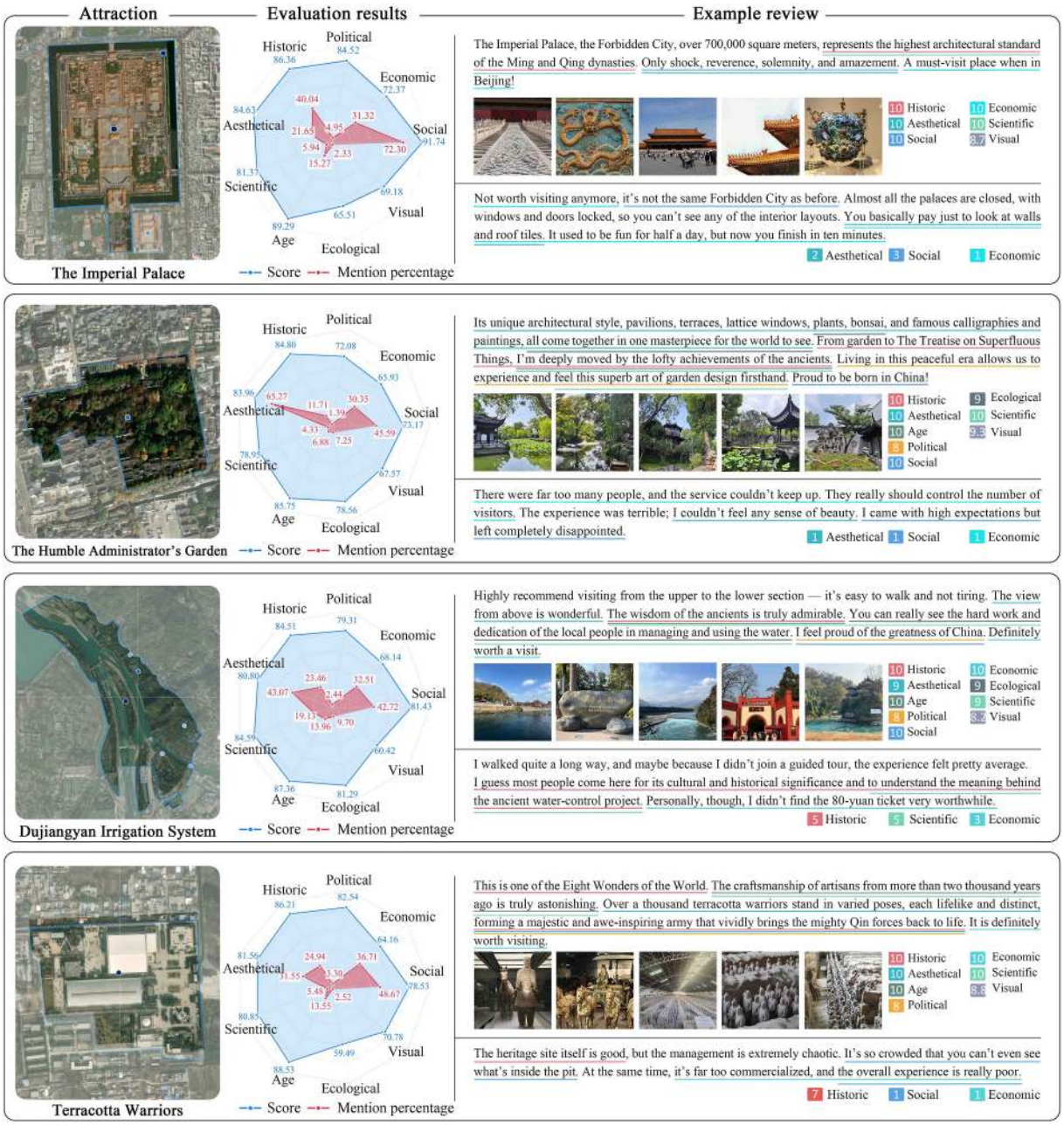


Fig. 6. Examples of quantified cultural values derived from reviews.

(40,008) and "value for money" (20,831), concentrating in high-tourism regions and traditional commercial streets such as the Bund (Shanghai) and Hongyadong (Chongqing).

Historic (28.28%) and age (7.69%) values share keywords like "history" and "architecture." However, historic value emphasizes educational and symbolic links to past events (e.g., "historical stories" (8273)), whereas age value highlights material endurance through "thousand-year" (4869) and "long-standing" (2641), with more high-high clusters in central and southwestern China. Scientific value (10.86%) centers on workmanship and technological skill, with "architecture" (5312) and "tour guide service" (4054) ranking highly, the latter indicating that professional interpretation is vital for communicating scientific meaning. Ecological value (18.38%) mainly reflects perceived environmental comfort and naturalness, with high-high clusters in eastern and southwestern China but notably absent around Beijing. Political value is least mentioned (4.13%), associated with nationhood and authority, with high-high clusters confined to explicitly revolutionary or state-symbolic sites such as Ruijin.

4.3.3. Temporal patterns

As shown in Fig. 10a, visitation volume increases during major travel periods, with clear peaks around February (Spring Festival), early May (Labor Day holiday), and early October (National Day Golden Week). In Fig. 10b, most cultural values show a concurrent decline during these high-visitation periods, suggesting that crowding and service overload may negatively influence the on-site experience.

4.4. Relationships between perceived attractiveness and visitation and explanatory variables

Among the tested models, XGBoost exhibited the most robust performance for both predictive tasks (Table 4) and was therefore selected for subsequent SHAP-based analyses. Importantly, incorporating city-level conditions consistently improved model fit for both perceived

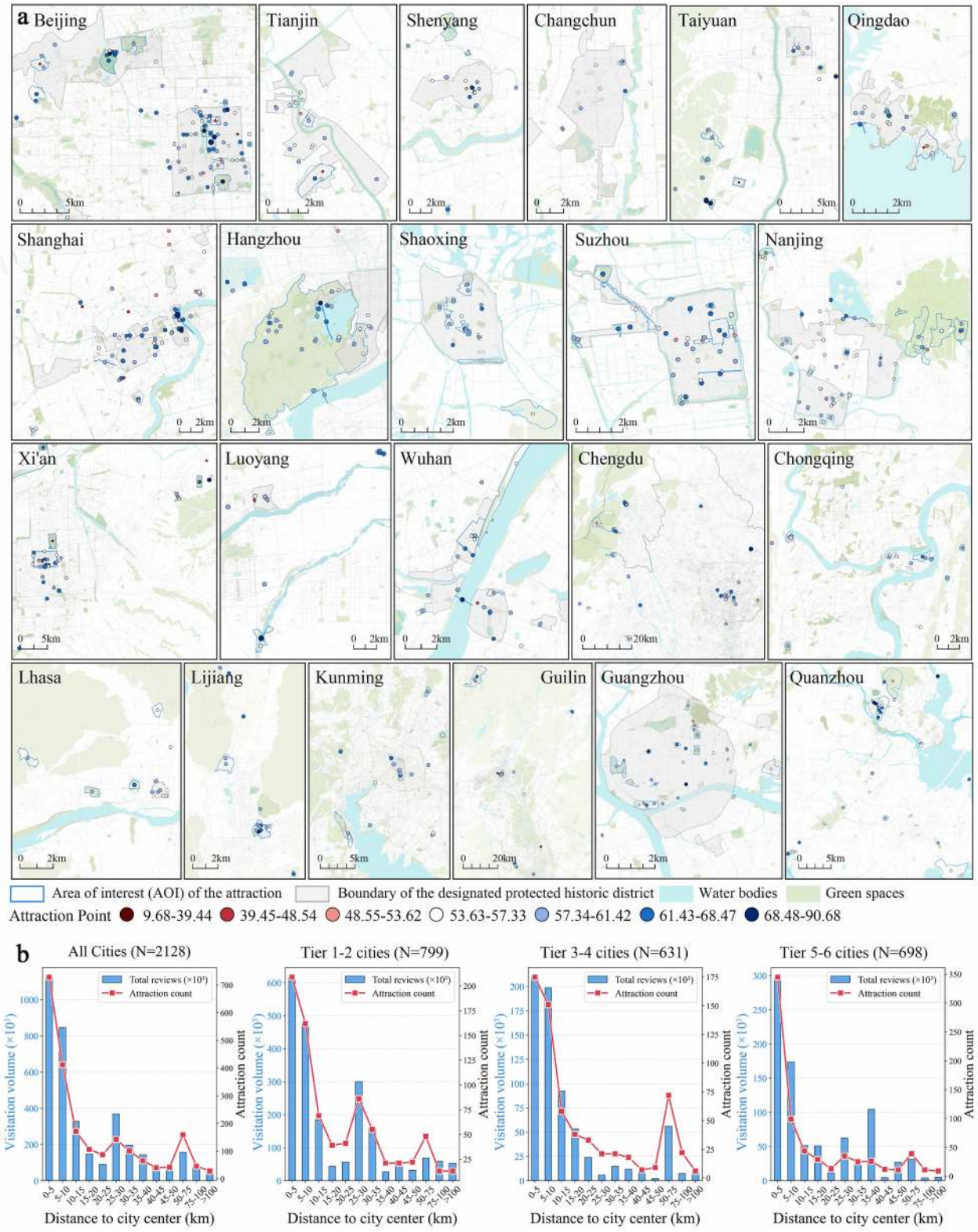


Fig. 7. Intra-city-scale spatial patterns: (a) maps of perceived attractiveness in representative cities; (b) distance-to-city-center gradients of attraction counts and visitation volumes.

attractiveness and visitation, yielding an average R^2 increase of 0.0529 (Appendix E). This improvement indicates that broader urban context captures between-city heterogeneity and supports more accurate modeling of perceived attractiveness and visitation outcomes.

4.4.1. *Influencing factors of perceived attractiveness*

Fig. 11 interprets the perceived-attractiveness model through a SHAP summary plot and feature-wise dependence curves to illustrate overall feature importance and nonlinear effects. Attraction attributes dominate the model. NKCRPU score and visual quality are the strongest

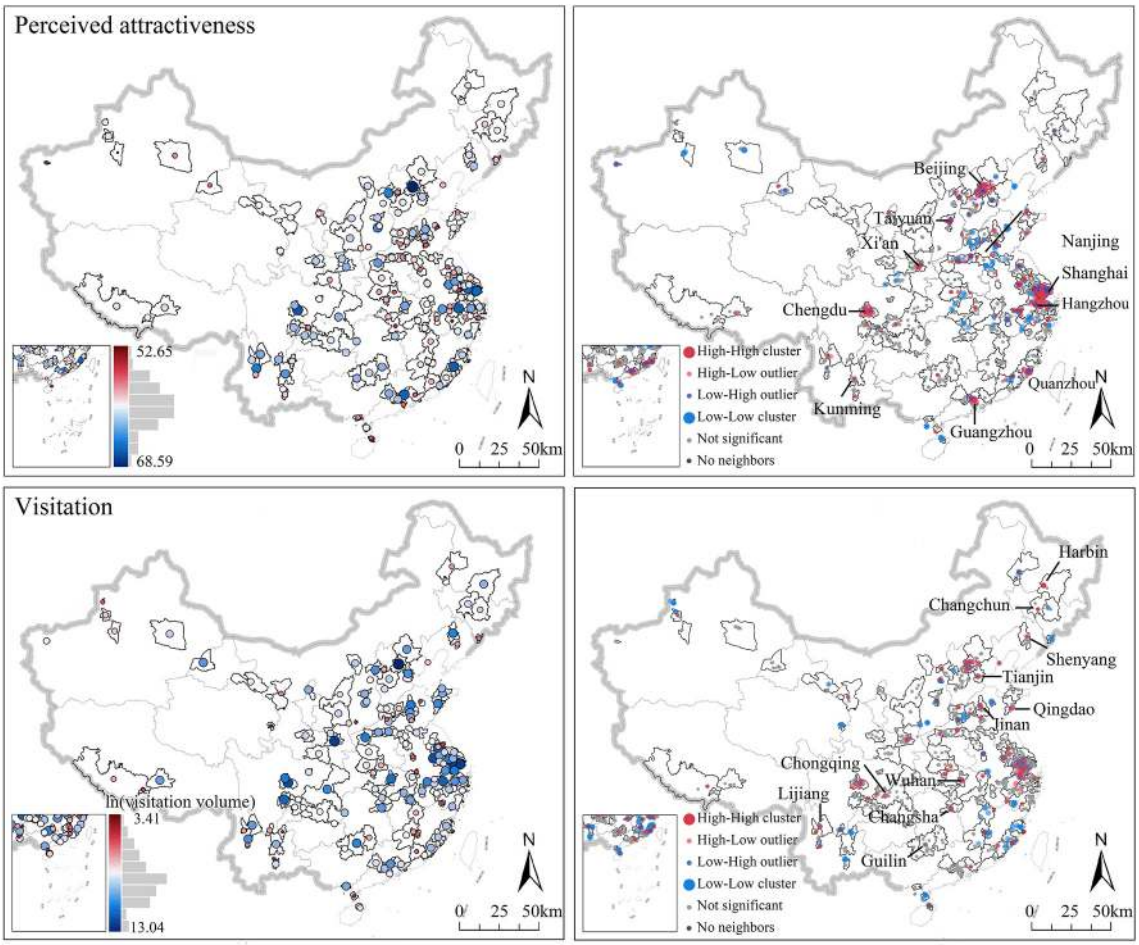


Fig. 8. Spatial distribution and LISA cluster maps for perceived attractiveness and visitation. Visitation volume was ln-transformed for visualization and local spatial association analysis to reduce the influence of the highly skewed distribution.

predictors, exhibiting broadly monotonic positive trends where higher NKCRPU scores and better visual aesthetics contribute positively to predicted perceived attractiveness. Conversely, ticket price and area contribute negatively, and both relationships flatten at higher values, with the effects weakening beyond about 100 yuan and 50 ha.

Supporting factors make secondary but still important contributions. Shorter distance to the city center and higher road and road intersection densities indicate better accessibility and local network connectivity, which are associated with higher perceived attractiveness. Activity and service intensity metrics, such as the densities of cultural facilities, tourist attractions, and retail, show positive contributions but exhibit saturation and even diminishing returns. Hotel density shows an overall negative effect, with a sharp marginal penalty as density increases from very low to moderate levels. Beyond an index of around 4, higher POI diversity is associated with lower perceived attractiveness. City-level tourism revenue also appears among the top 15 variables and shows a modest positive contribution.

4.4.2. *Influencing factors of visitation*

For visitation (Fig. 12), attraction attributes remain dominant. Ticket price, A-level score, area, visual quality, and NKCRPU score all contribute positively, with clear nonlinearities: the effect of ticket price largely plateaus around 100 yuan, the marginal gains of increasing area diminish beyond roughly 100 ha, and visual quality becomes distinctly positive once scores exceed about 40. Among perceived cultural values, social and historic values act as high-end amplifiers, showing limited impact at lower levels but rising sharply once scores exceed the 55-60 range. Economic and scientific values also contribute positively, saturating at scores of approximately 60 and 40, respectively. In contrast, political value is associated with lower visitation overall. Ecological and age values do not appear among the top twenty predictors, suggesting limited explanatory power.

Regarding supporting factors, tourist attraction density contributes positively before saturating, whereas hotel density initially contributes positively to visitation but turns negative at higher levels, suggesting penalties for excessive concentration. Higher road intersection density and proximity to airports are associated with higher visitation, highlighting the role of transport convenience. Street-view-based beautiful and lively scores also contribute positively. Finally, city-level conditions add a smaller but consistent uplift. Tourism revenue and GDP per capita show positive associations, while population shows a comparatively weak positive contribution concentrated in the largest cities.

5. Discussion

5.1. *Spatial and temporal patterns of perceived attractiveness and visitation*

Spatially, both attraction counts and visitation volumes decay with distance to the city center and exhibit a long-tail structure at the city and attraction scales, with a small number of cities and a few top attractions within each city accounting for most of the total. These patterns reinforce a long-standing finding that visitation concentrates in well-connected central districts and iconic attractions (Li et al., 2018; Paül i Agustí, 2018). This polarization reflects the organizational logic of attraction systems, where identifiable flagship attractions are promoted

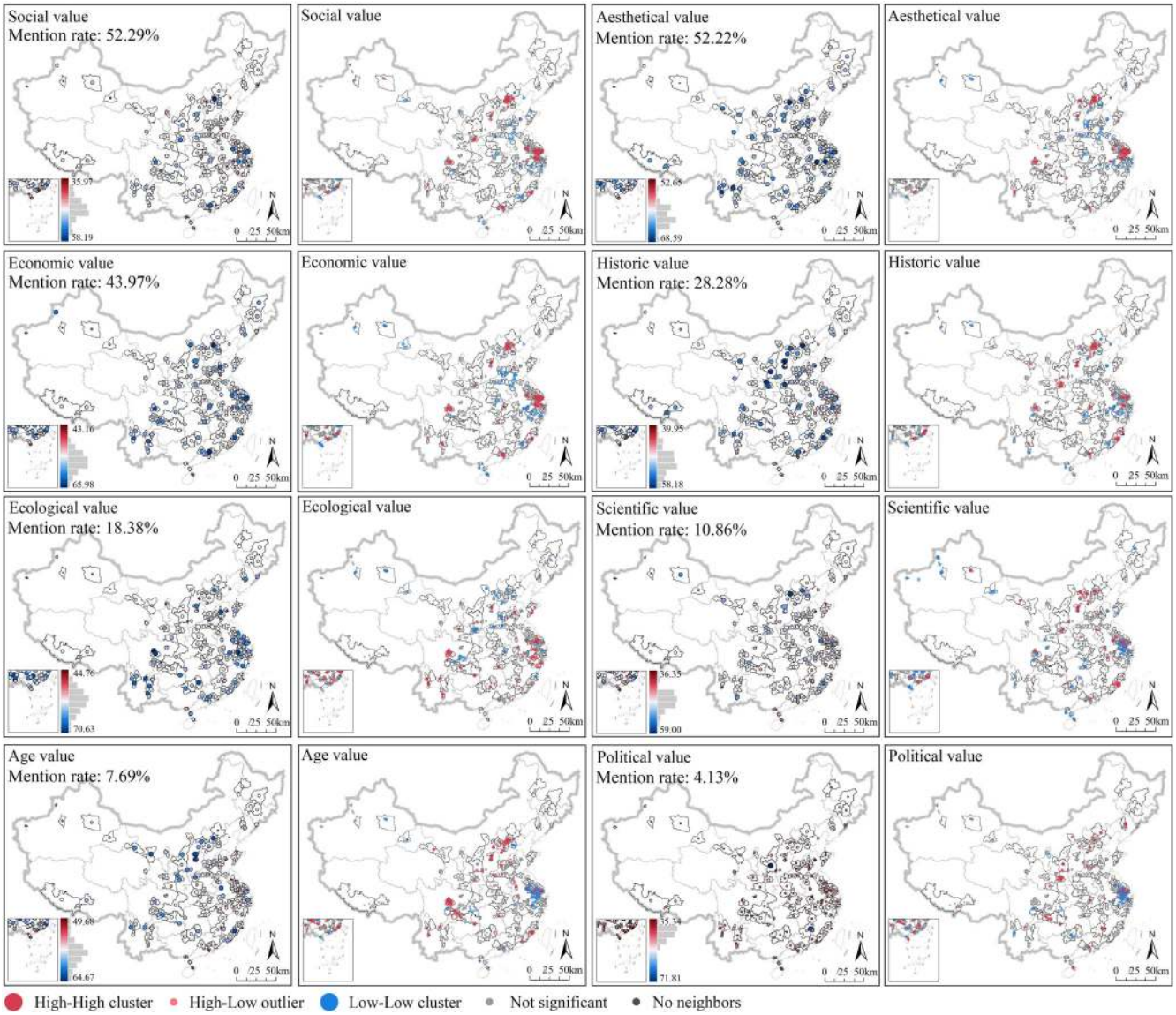


Fig. 9. Spatial distribution and LISA cluster maps for the eight cultural values.

as primary destination markers, funneling demand disproportionately to specific locations while leaving others underutilized (Lew, 1987; Cheng et al., 2025). To raise overall system performance, authorities should maintain flagship attractions while facilitating visitor dispersal through better connections to secondary attractions such as improved public transport and last-mile access and integrated thematic routes that encourage multi-stop itineraries. Additionally, the higher perceived attractiveness and visitation inside protected historic districts further supports Lew (1987)'s organizational perspective, wherein these districts can be understood as integrated destination-attraction complexes that bundle multiple attractions, activities, and supporting services into a coherent spatial unit, thereby concentrating tourist attention. Consequently, authorities should continue to integrate historic district protection with more coherent tourism system planning. Furthermore, the success of some peripheral destination-type attractions indicates that location is not a deterministic constraint; instead, certain attractions possess sufficient pull to serve as primary trip anchors, overcoming distance decay to actively shape destination choices (Leiper, 1990).

At the inter-city scale, perceived attractiveness and visitation, together with the eight cultural values, exhibit positive spatial autocorrelation, and LISA reveals a stable multi-core structure with three dominant high-high clusters around Beijing, Shanghai, and Chengdu, alongside smaller clusters surrounding regional hubs. Importantly, these cores are not always uniform. For example, ecological value does not form a high-high cluster around Beijing, and age value is comparatively weaker around Shanghai, indicating value-specific geographies.

Temporally, visitation peaks during major holidays, consistent with UGC-based research on urban open spaces (Zhao et al., 2025; Li et al., 2026). Our value-based measurement further reveals that most perceived cultural-value scores decline during these peak periods. This aligns with tourism carrying-capacity arguments (Theobald, 2012) and broader destination-impact perspectives highlighting that overcrowding and strained conditions threaten the visitor experience (Crouch & Ritchie, 1999).

5.2. *Influencing factors of perceived attractiveness and visitation*

Attraction attributes contribute most significantly to both perceived attractiveness and visitation (MacCannell, 1976; Crouch & Ritchie, 1999; Ferrario, 1979; Formica & Uysal, 2006; Leiper, 1990; Lew, 1987), and our analysis reveals several noteworthy patterns. First, destination pricing and reputation theory argues that when tourists

Table 3
Value-specific keywords and frequencies derived from reviews.

Social	Aesthetical	Economic	Historic	Age	Scientific	Ecological	Political
worth visiting (30,507)	beautiful (42,155)	worth visiting (40,008)	history (44,705)	history (15,494)	architecture (5,312)	environment (5,221)	history (2,080)
good (27,899)	scenery (37,187)	good (24,551)	culture (13,473)	architecture (8,652)	tour guide service (4,054)	good (2,607)	China (2,001)
history (26,967)	architecture (27,532)	ticket (23,383)	architecture (11,352)	first built (4,927)	craftsmanship (2,694)	fresh air (2,294)	culture (1,371)
tour guide service (20,584)	landscape (20,431)	value for money (20,831)	learning value (10,831)	thousand-year (4,869)	design (2,101)	pleasant environment (1,777)	education (1,246)
culture (17,017)	scenic area (19,977)	recommend (19,367)	China (10,580)	preservation (2,921)	China (1,644)	park (1,745)	royal (1,163)
experience (15,501)	garden (9,891)	commercial (13,683)	built in (historical period) (10,131)	historic site (2,766)	ancient wisdom (1,494)	nature (1,588)	government (1,090)
children (15,240)	magnificent (9,473)	free of charge (12,355)	tour guide service (10,045)	old (2,703)	art (1,476)	refreshing (1,447)	emperor (1,037)
tour guide (12,508)	photo-worthy (9,082)	tourists (8,348)	ancient times (9,538)	long-standing (2,641)	structure (1,431)	scenery (1,318)	patriotism (1,028)
emotional response (12,411)	exquisite (8,501)	affordable (8,178)	historical stories (8,273)	Qing dynasty (2,548)	exquisite (1,399)	architecture (1,281)	revolution (892)
learning (11,785)	awe-inspiring (8,488)	ticket price (7,739)	ruins (4,670)	ruins (2,491)	carving (1,380)	landscape (1,132)	commemoration (851)

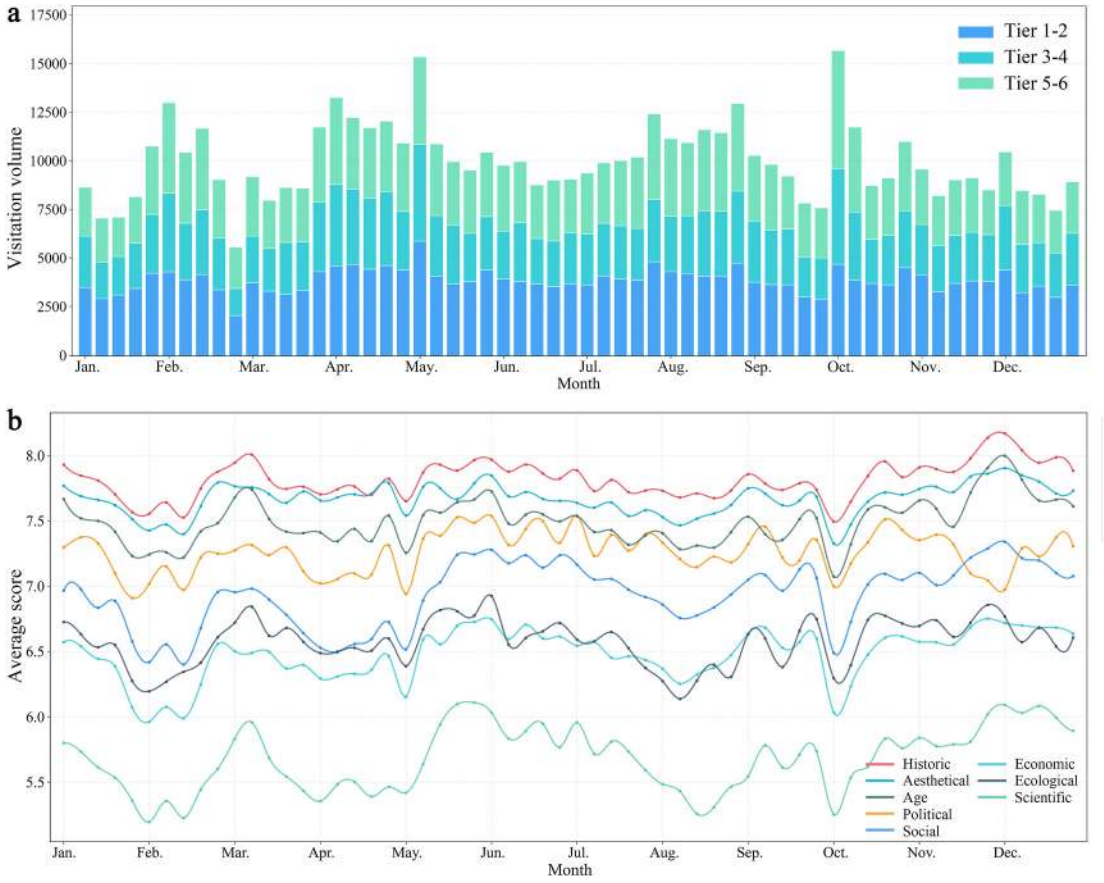


Fig. 10. Temporal fluctuations of visitation volume and eight cultural value scores.

cannot easily assess attractions' quality in advance, they rely on visible signals such as official designations or prices to guide their choices (Keane, 1997). While the WHS designation is widely regarded as a value-adding asset for tourism development (Koufodontis & Gaki, 2022), it does not emerge as a key predictor in our models, likely because only 14.10% of sampled attractions fall within WHS zones. By contrast, two China-specific recognition systems show clearly divergent roles. The NKCRPU score most strongly predicts perceived attractiveness, whereas the A-level score is among the top predictors of visitation. In China, the A-level rating (5A/4A/3A) is a tourism-facing certification highly visible in destination marketing. It consolidates multiple dimensions of service readiness, such as facilities, management, and reception capacity, into a comparable hierarchy, making it more likely to translate into actual visitation. Conversely, NKCRPU designations are

Table 4
Performance comparison of candidate models for predicting perceived attractiveness and visitation.

Model	R^2	RMSE	MAE
Y = perceived attractiveness (range: 8.61-89.74)			
XGBoost	**0.6264**	**4.2586**	**3.0049**
Gradient Boosting	0.6081	4.3621	3.0548
Random Forest	0.6078	4.3634	3.1031
Support Vector Regression	0.6037	4.3863	3.2310
Ridge Regression	0.5569	4.4565	3.2511
Lasso Regression	0.5556	4.4570	3.2581
Y = visitation (ln-transformed visitation volume; range: 3.91-12.10)			
XGBoost	**0.7589**	**0.0724**	**0.0549**
Gradient Boosting	0.7495	0.0738	0.0563
Random Forest	0.6878	0.0824	0.0626
Support Vector Regression	0.6640	0.0855	0.0638
Ridge Regression	0.5202	0.1021	0.0780
Lasso Regression	0.5175	0.1024	0.0780

awarded to heritage assets for their exceptional cultural significance rather than for tourism readiness. Their strong association with perceived attractiveness indicates that truly unique heritage resources contribute directly to tourists' experiential evaluations, a finding consistent with questionnaire-based evidence (Vengesayi et al., 2009). Consequently, management strategies should clarify each attraction's unique qualities and communicate them through high-quality interpretation (Reitsamer et al., 2016) and targeted marketing (Formica & Uysal, 2006, Du, & LiechtySantosPark, 2022).

A second key finding concerns price. Consistent with signaling logic (Keane, 1997), ticket price contributes positively to visitation, suggesting that higher prices can communicate higher must-see status, reputation, or expected quality and thus draw visitation. However, higher prices can depress perceived attractiveness, consistent with findings that higher fees are more likely to trigger value-for-money complaints and expectation-performance mismatches (Riswanto et al., 2025). Additionally, economic value was a central theme in 43.97% of reviews, primarily regarding ticket pricing and perceived worth. These findings underscore the need for optimized pricing strategies that improve perceived fairness. Ensuring the cost is justified by the experience is essential for maintaining positive evaluations and encouraging visitation rather than deterring it. Additionally, our results show that visual quality positively influences both perceived attractiveness and visitation. Streetscape metrics such as lower sky proportion, higher greenery, perceived beauty and liveliness are also beneficial. These suggest that high-quality environments both within and surrounding the attraction are associated with better visitor experiences and increased visitation (Rui et al., 2025; Zhao et al., 2025).

For visitation, perceived social, economic, scientific, historic, and aesthetical values emerge as important positive predictors, suggesting that positive tourist perceptions are instrumental in driving visitation. For social and historic values, visitation is relatively flat at low-to-moderate perception levels but rises sharply once scores reach an upper-tier threshold. This implies that small improvements in these dimensions may not increase visits unless they are strong enough to push

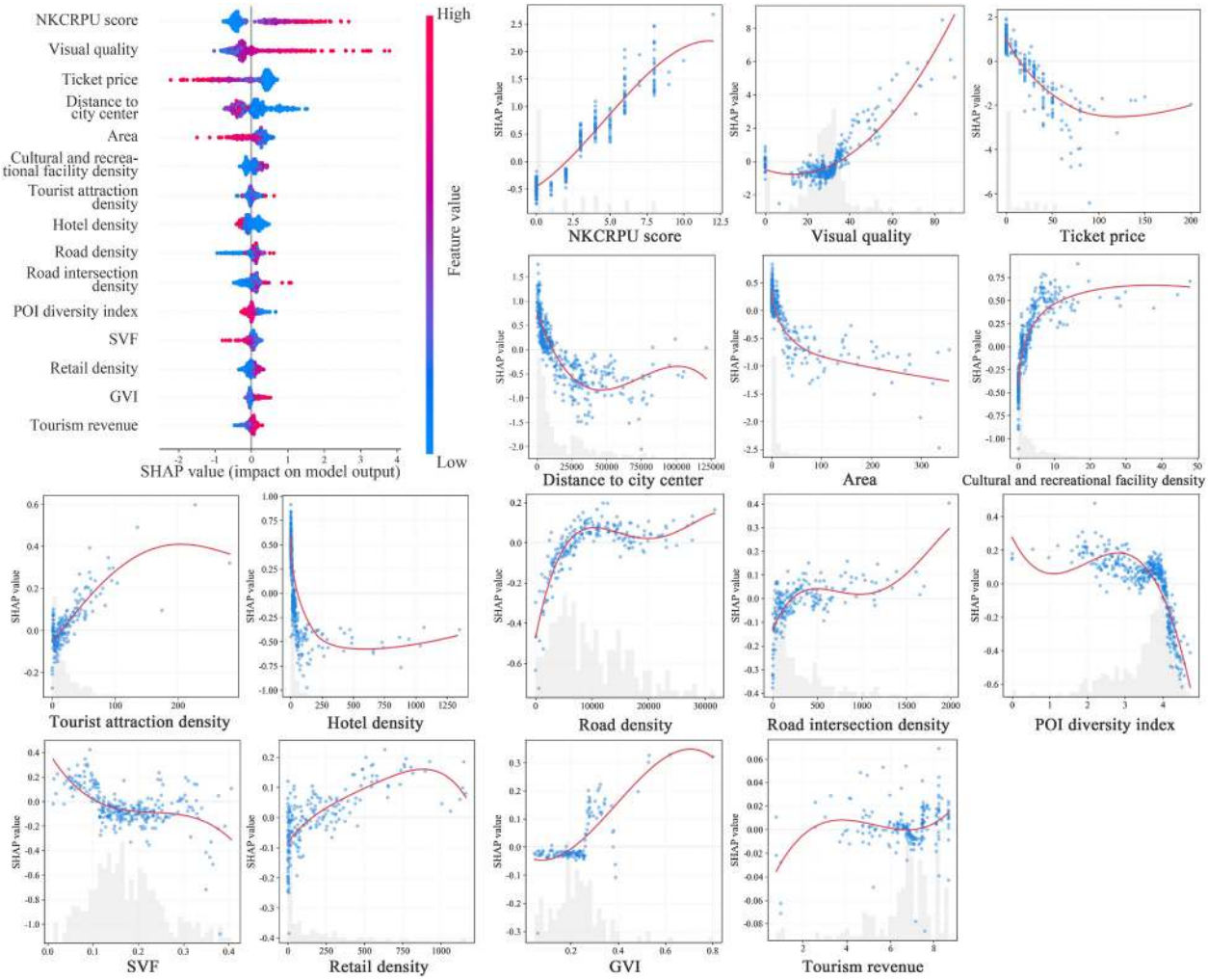


Fig. 11. Feature importance and nonlinear effects of key variables on perceived attractiveness.

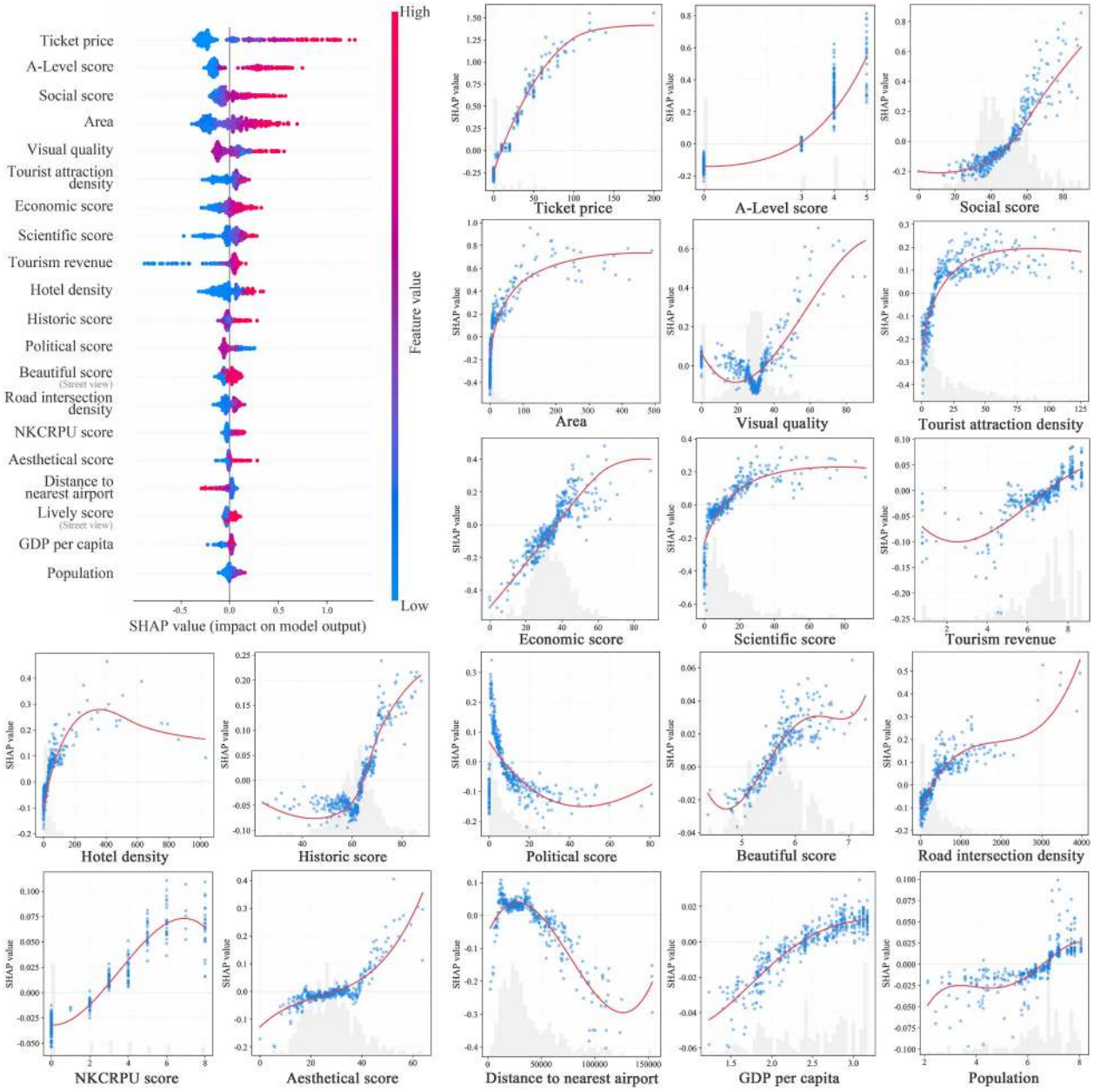


Fig. 12. Feature importance and nonlinear effects of key variables on visitation.

the attraction into a clearly differentiated, high-perception bracket. Regarding scientific value, most attractions receive low scores and are seldom perceived by the public. However, attractions that achieve even modest scientific-value scores show a marked increase in visitation, suggesting that clearly communicating technical or educational content can serve as a strong differentiator. At the same time, the negative association with political value and the weak explanatory power of ecological and age values indicate that not all perceived meanings convert equally into footfall. Visitation growth depends more on achieving standout, broadly appealing narratives and experiences than on uniformly raising every value dimension.

Finally, supporting factors constitute the essential foundation of the tourism industry, acting as the necessary prerequisite for core resources to be accessed and experienced (Ferrario, 1979; Crouch & Ritchie, 1999; Vengesayi et al., 2009; Reitsamer et al., 2016). In our results, accessibility-related variables such as distance to the city center, road density, road intersection density, and distance to the nearest airport emerge as important predictors of attractiveness. Moreover, the density of tourist attractions increases both perceived attractiveness and visitation, supporting the notion of cluster-based destination competitiveness where individual attractions benefit from the collective pull and synergy of the surrounding region (Crouch & Ritchie, 1999; Giambona & Grassini, 2020; Koufodontis & Gaki, 2022). However, amenity structure is not uniformly beneficial. While accommodation density is a key component of destination support services (Vengesayi, Mavondo et al., 2009), it exhibits clear threshold effects in our results, and excessive accommodation concentration may also reflect overdevelopment pressures that undermine experiential quality (Gunn & Var, 2002). Similarly, POI diversity becomes clearly detrimental beyond an index of around 4, suggesting that excessive functional mixing may dilute the attraction focus and reduce the clarity of the tourism offer. In areas with an oversupply of amenities, the focus should shift from expansion to quality upgrades and better spatial alignment with visitor flows. City-level conditions add a smaller but consistent uplift, with higher tourism revenue and GDP per capita associated with higher attractiveness.

5.3. Theoretical and methodological contributions

First, this study advances tourism attractiveness research by translating the classic attraction-system perspective (MacCannell, 1976; Ferrario, 1979; Lew, 1987; Leiper, 1990; Crouch & Ritchie, 1999; Formica & Uysal, 2006; Vengesayi et al., 2009) into a tourist-centered, multi-scale analytical framework. This framework quantifies perceived attractiveness and visitation at the attraction scale, aggregates them to the city scale, and links both outcomes to attraction attributes, supporting factors, and city-level conditions. It both reveals the spatial and temporal patterns of perceived attractiveness and visitation across attractions and cities and examines how these outcomes are jointly produced within large heritage-attraction systems. In doing so, our study addresses three gaps in the literature: large-scale analyses often rely on aggregated indicators that obscure attraction-specific perceptions (Formica & Uysal, 2006; Giambona & Grassini, 2020); attraction-level studies of tourist perceptions have been largely confined to small-sample surveys (Vengesayi et al., 2009; Reitsamer et al., 2016); and diverse explanatory variables have rarely been integrated within a coherent framework.

Second, this study advances the conceptualization of perceived attractiveness in tourism research. Prior tourism-perception studies often rely on ratings or overall sentiment scores (Bi, & ZhuHan, 2024; Huang, & ShiLiWang, 2025; Li et al., 2026), which may obscure the multidimensional meanings through which tourists evaluate attractions. By measuring perceived attractiveness through eight cultural values, this study conceptualizes tourist perception of heritage as a value-specific and interpretable construct. The findings further show that these values do not translate into visitation in the same way. Social, historic, economic, scientific, and aesthetical values contribute positively to visitation, whereas political value is negatively associated with visitation and ecological and age values show limited explanatory power. This relationship is also nonlinear for some dimensions, as social and historic values convert into visitation mainly once they pass an upper-tier level rather than rising steadily across the whole range. Together these patterns indicate that the perception-to-visitation relationship is value-specific and uneven rather than uniform. The attraction-scale value profiles also help narrow the gap between expert-defined and tourist-perceived heritage values, thereby supporting more evidence-based decision-making (Ginzarly et al., 2019).

Third, our findings imply that perceived attractiveness and visitation operate as related but partially independent constructs shaped by different factors, and we further specify how individual factors drive each outcome. For example, national heritage significance, represented by NKCRPU status, is most strongly associated with perceived attractiveness, whereas tourism-facing A-level certification and ticket price are more strongly associated with visitation.

Methodologically, this study demonstrates how theory-guided AI pipelines can be used to quantify perceptions and visual quality from multi-modal UGC using LLMs and VLMs, without reliance on extensive domain-specific labeled datasets or costly supervised training (Koufodontis & Gaki, 2022; Zhao et al., 2024; Rui et al., 2025; Zhao et al., 2025). The LLM component transforms reviews into structured cultural value measures through extraction and scoring, while design choices such as rolling-window prompt memory and verbatim evidence checking improve stability and reduce hallucination risk. The VLM component offers a scalable proxy for the visual quality of attractions that is otherwise difficult to measure consistently across thousands of attractions. Overall, these approaches demonstrate replicable, low-barrier pathways for tourism and urban studies to move beyond coarse evaluative proxies toward interpretable, theory-aligned measures of tourist perception.

5.4. Policy and practical implications

Our framework yields three classes of policy-relevant diagnostic outputs: attraction-scale value profiles (Fig. 6), intra-city and inter-city benchmarking (Figs. 7–9), and city- and attraction-level typologies (Fig. 13, Fig. S6 and S7). Together with the explanatory insights from the identified influencing factors, these outputs help diagnose whether performance bottlenecks stem from experience-quality deficits or limited visitation, while the value profiles and influencing factors further guide the strengthening of underperforming values and the branding of dominant ones.

Taking the city typology in Fig. 13a as an example, cities high in both perceived attractiveness and visitation should prioritize upgrading weaker attractions while maintaining capacity controls and experience-quality safeguards during peak periods. Low-perception but high-visitation cities should focus on quality recovery, including crowding mitigation, service and interpretation upgrades, and better price-value alignment. High-perception but low-visitation cities require demand activation through better accessibility and supporting factors, itinerary integration, and targeted information and marketing to convert positive evaluations into trips. Cities low in both metrics need selective upgrading and clearer market positioning, supported by benchmarking against reference cities with similar GDP, population, and heritage endowments but stronger performance. At the attraction scale, the same logic applies through quadrant analysis (Fig. 13b).

The spatial classification map for Beijing (Fig. 13c) further translates this typology into a planning map by showing where different categories of attractions are located. This enables planners to identify where visitor pressure should be dispersed, where under-visited but well-perceived attractions could be integrated into routes, and where experience-quality improvement is most urgent. For example, Nanluoguxiang appears as a highly visited attraction with comparatively weaker perceptions. While it benefits from excellent accessibility and consistently draws tourists, the on-site experience often falls short of expectations. The UGC-based value portfolio highlights recurring deficits such as “missing the local lived-in vibe” and “way too commercial” (Fig. 13d), helping pinpoint more targeted enhancement priorities. Continuous monitoring of value-specific review signals can then verify whether interventions are improving the targeted weak aspects.

5.5. Limitations and future research

First, both UGC and AI-based evaluations contain inherent bias and may not fully capture the tourist experience (Zhao et al., 2024; Luo et al., 2025). Specifically, LLMs and VLMs may reflect internal biases from their massive training corpora, and the predictive performance reported in this study was evaluated on a relatively small test set. Consequently, further verification and large-scale validation are encouraged in future research. Additionally, although social media participation is expanding, platform users remain demographically selective, with persistent gaps across age, income, and education that can skew both review content and inferred preferences (Li et al., 2018; Mor, & DalyotRam, 2023). Nonetheless, our study demonstrates a transferable approach to measuring attractiveness using UGC. Second, the explanatory variables considered are not comprehensive, as unobserved factors such as marketing campaigns, temporary closures, and special events may confound associations (Formica & Uysal, 2006). Consequently, future work should combine UGC-based measures with surveys and supplementary operational indicators. Third, our findings may be context-specific and limited in generalizability for cities not centered on heritage attractions. Finally, the SHAP dependence plots should be interpreted as model-based tendencies rather than deterministic effects. For example, the negative pattern between ticket price and perceived attractiveness does not imply that higher prices will necessarily reduce perceived attractiveness in every case.

6. Conclusions

This study develops a tourist-centered, theoretical-to-applied

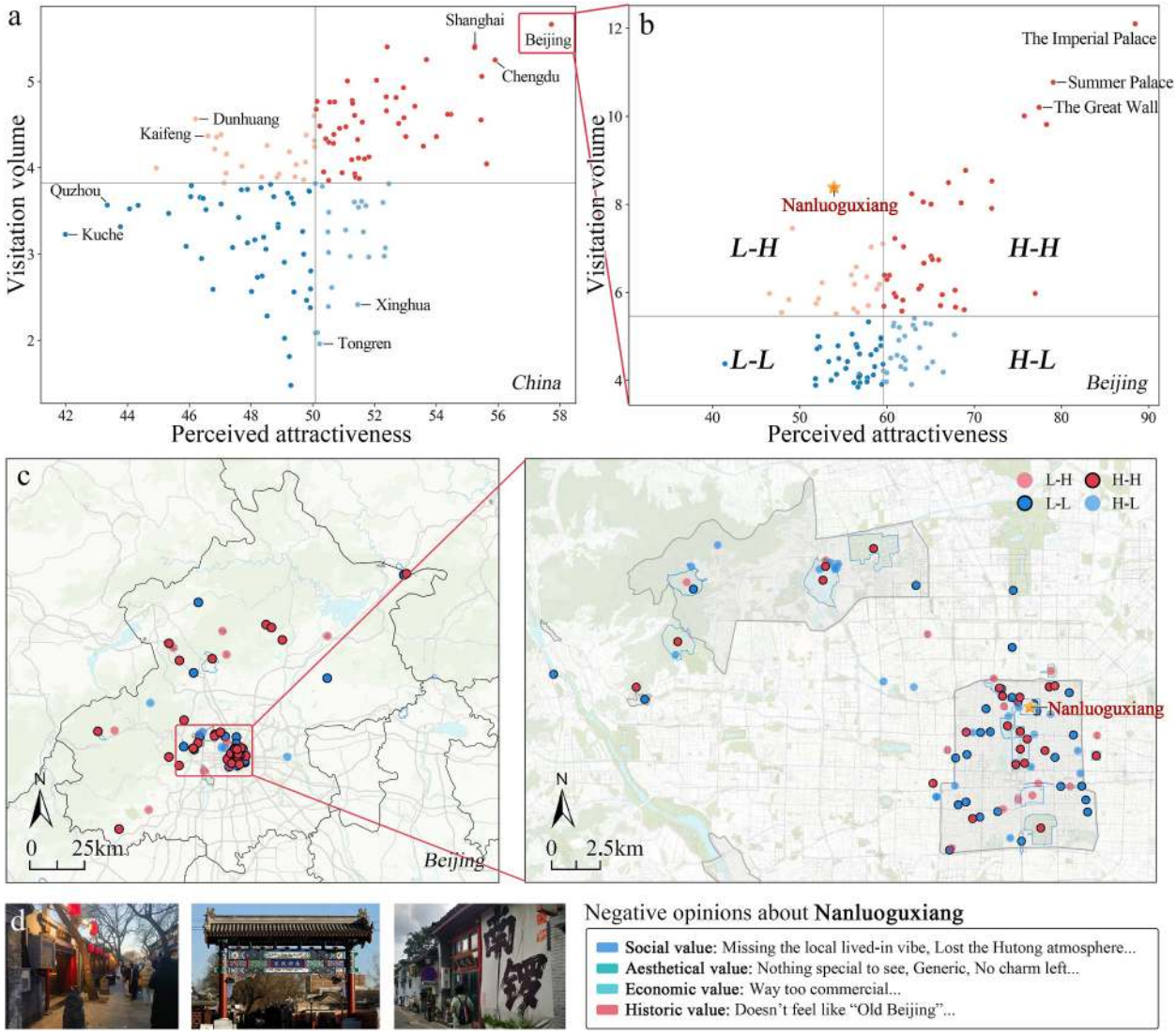


Fig. 13. Multi-scale perceived attractiveness-visitation typology and diagnostic application. (a) City-level quadrant analysis of perceived attractiveness and visitation across China. (b) Attraction-level quadrant analysis in Beijing. (c) Spatial distribution of Beijing attractions by quadrant category, with an inset highlighting the central urban area. (d) Illustrative case of Nanluoguxiang, including representative street scenes and summarized negative opinions.

framework that measures perceived attractiveness and visitation at multiple scales and connects them to explanatory variables. Our results reveal several key insights.

Spatially, both attraction counts and visitation volumes exhibit a clear distance-to-city-center decay and a highly polarized long-tail structure. Heritage attractions within protected historic districts demonstrate higher perceived attractiveness and visitation. At the inter-city scale, attractiveness exhibits a multi-core national pattern, with persistent high-performance clusters centered around Beijing, Shanghai, and Chengdu. Temporally, major holidays substantially increase visitation but are associated with lower perceived attractiveness, indicating a tension between visitation and experience quality.

In terms of influencing factors, attraction attributes dominate both metrics. Perceived attractiveness is most strongly enhanced by national heritage status (NKCRPU) and superior visual quality, while higher ticket prices tend to undermine tourists' evaluations. Visitation is positively associated with price signaling and A-level certifications, and is further strengthened by higher perceived social, economic, scientific, historic and aesthetical cultural values. Conversely, political value contributes negatively, while ecological and age values show limited explanatory power for visitation. Additionally, supporting factors, including accessibility and facility densities, have secondary but still important influences. Beyond individual sites, both perceived attractiveness and visitation benefit from spatial agglomeration, as dense clusters of nearby attractions and cultural facilities generate collective pull and regional synergy. Finally, high-quality environments both within attractions and in their surrounding areas contribute to improved visitor experiences and higher visitation.

From an applied perspective, our framework produces attraction-scale, multidimensional value profiles with traceable textual evidence, along with multi-scale benchmarking maps and city- and attraction-level typologies. These tools diagnose whether underperformance stems from perception deficits, limited conversion from perception to visitation, or resource constraints, supporting targeted strategies for sustainable tourism management and urban planning. More broadly, the LLM-based approach demonstrates a replicable pathway for extending perceived attractiveness research beyond coarse proxies, enabling interpretable, theory-aligned measurements of tourist perception that can be applied to other attraction types and urban spaces.

CRediT authorship contribution statement

Xukai Zhao: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **He Huang:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision. **Nan Bai:** Conceptualization, Investigation, Methodology, Project administration, Validation, Writing – review & editing. **Yuxuan Chen:** Formal analysis, Methodology, Validation. **Yecheng Zhang:** Conceptualization, Methodology, Validation. **Miao Wang:** Conceptualization, Writing – original draft. **Yuxing Lu:** Conceptualization, Investigation, Methodology, Writing – review & editing.

Impact statement

This study advances tourism-management research by developing a tourist-centered framework that first measures perceived attractiveness and visitation, and then examines how attraction attributes, supporting factors, and city-level conditions shape perceived attractiveness and how perceived attractiveness together with these variables shapes visitation. Analyzing 610,136 reviews across 2128 heritage attractions in 143 Chinese cities, the study combines large language models, review-volume-based visitation measurement, and interpretable machine learning to reveal multi-scale spatiotemporal patterns and the factors influencing them. Its practical contribution lies in generating diagnostic benchmarking and attraction-scale value profiles that help heritage managers and policymakers identify underperforming attractions and cities and prioritize targeted improvements.

Declaration of AI and AI-assisted technologies

AI-assisted tools were used to support language editing and code development during manuscript preparation. All AI-generated text, code, and outputs were reviewed, revised, and verified by the authors, who take full responsibility for the integrity and accuracy of the final manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tourman.2026.105515.

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